Chapter1 Basis Expansion

In order to derive the two-body matrix elements of the two-body currents operator, we need to transform the absolute coordinate system representing the single-particle wave function into the Jacobi coordinate system (Moshinsky Transformation). However, the single-particle wave function obtained from our program is a function related to the coordinate space, so it is necessary to use the basis expansion method to project it onto the spherical harmonic oscillator basis.

For two-particle systerms, define the $\psi_{n_a l_a j_a m_a \tau_a}(\boldsymbol{r}) = \sum\limits_{nljm} c_{nljm}^{n_a l_a j_a m_a \tau_a} \phi_{nljm}(\boldsymbol{r}), \phi_{nljm}(\boldsymbol{r})$ is the Harmonic Oscillator Wave Function.

$$|\psi\rangle = \sum_i |\phi_i\rangle\langle\phi_i| \cdot |\psi\rangle \tag{1-1}$$

$$= \sum_i \int_{\boldsymbol{r}'}\int_{\boldsymbol{r}''} |\boldsymbol{r}''\rangle\langle\boldsymbol{r}''|\phi_i\rangle\langle\phi_i|\boldsymbol{r}'\rangle\langle\boldsymbol{r}'|\psi\rangle \tag{1-2}$$

$$= \sum_i \int_{\boldsymbol{r}'}\int_{\boldsymbol{r}''} |\boldsymbol{r}''\rangle\phi_i(\boldsymbol{r}'')\phi_i^\dagger(\boldsymbol{r}')\psi(\boldsymbol{r}') \tag{1-3}$$

Left multiply by $\langle\boldsymbol{r}|$ in above equation

$$\langle\boldsymbol{r}|\psi\rangle = \sum_i \int_{\boldsymbol{r}'}\int_{\boldsymbol{r}''} \langle\boldsymbol{r}|\boldsymbol{r}''\rangle\phi_i(\boldsymbol{r}'')\phi_i^\dagger(\boldsymbol{r}')\psi(\boldsymbol{r}') \tag{1-4}$$

so we can get

$$\psi_j(\boldsymbol{r}) = \sum_i \int_{\boldsymbol{r}'} \phi_i^\dagger(\boldsymbol{r}')\psi_j(\boldsymbol{r}')\phi_i(\boldsymbol{r}) = \psi_{n_j l_j j_j m_j \tau_j}(\boldsymbol{r}) = \sum_i c_i^j \phi_i(\boldsymbol{r}) \tag{1-5}$$

$$= \sum_{n_i l_i j_i m_i} \int_{\boldsymbol{r}'} \phi_{n_i l_i j_i m_i}^\dagger(\boldsymbol{r}')\psi_{n_j l_j j_j m_j \tau_j}(\boldsymbol{r}')\phi_{n_i l_i j_i m_i}(\boldsymbol{r}) = \sum_{n_i l_i j_i m_i} c_{n_i l_i j_i m_i}^{n_j l_j j_j m_j \tau_j} \phi_{n_i l_i j_i m_i}(\boldsymbol{r}) \tag{1-6}$$

$$= \sum_{n_i l_i j_i m_i} \int_{\boldsymbol{r}'} \left(\eta_i g_{n_i l_i}(r') \sum_{m_{l_i} m_{s_i}} C_{l_i m_{l_i} \frac{1}{2} m_{s_i}}^{j_i m_i} Y_{l_i m_{l_i}}(\hat{\boldsymbol{r}}')\chi_{\frac{1}{2}m_{s_i}}\chi_{\frac{1}{2}m_{\tau_i}}\right)^\dagger$$

$$\cdot\left(\eta_j u_{n_j l_j \tau_j}(r') \sum_{m_{l_j} m_{s_j}} C_{l_j m_{l_j} \frac{1}{2} m_{s_j}}^{j_j m_j} Y_{l_j m_{l_j}}(\hat{\boldsymbol{r}}')\chi_{\frac{1}{2}m_{s_j}}\chi_{\frac{1}{2}m_{\tau_j}}\right) d\boldsymbol{r}'\phi_{n_i l_i j_i m_i}(\boldsymbol{r}) \tag{1-7}$$

$$= \sum_{n_i l_i j_i m_i} \sum_{m_{l_i} m_{s_i} m_{l_j} m_{s_j}} \eta_i^*\eta_j C_{l_i m_{l_i} \frac{1}{2} m_{s_i}}^{j_i m_i} C_{l_j m_{l_j} \frac{1}{2} m_{s_j}}^{j_j m_j} \int_{r'} g_{n_i l_i}^*(r') u_{n_j l_j \tau_j}(r') r'^2 dr'$$

$$\int_{\hat{\boldsymbol{r}}'} Y_{l_i m_{l_i}}^*(\hat{\boldsymbol{r}}')Y_{l_j m_{l_j}}(\hat{\boldsymbol{r}}')d\hat{\boldsymbol{r}}'\chi_{\frac{1}{2}m_{s_i}}^\dagger\chi_{\frac{1}{2}m_{s_j}}\chi_{\frac{1}{2}m_{\tau_i}}^\dagger\chi_{\frac{1}{2}m_{\tau_j}}\phi_{n_i l_i j_i m_i}(\boldsymbol{r}) \tag{1-8}$$

$$= \sum_{n_i l_i j_i m_i} \sum_{m_{l_i} m_{s_i} m_{l_j} m_{s_j}} \eta_i^* \eta_j C^{j_i m_i}_{l_i m_{l_i} \frac{1}{2} m_{s_i}} C^{j_j m_j}_{l_j m_{l_j} \frac{1}{2} m_{s_j}} \int_{r'} g^*_{n_i l_i}(r') u_{n_j l_j \tau_j}(r') r'^2 dr' \delta_{l_i l_j} \delta_{m_{l_i} m_{l_j}} \delta_{m_{s_i} m_{s_j}} \delta_{m_{\tau_i} m_{\tau_j}} \phi_{n_i l_i j_i m_i}(\boldsymbol{r}) \tag{1-9}$$

$$= \sum_{n_i j_i m_i} \sum_{m_{l_i} m_{s_i}} |\eta_j|^2 C^{j_i m_i}_{l_j m_{l_i} \frac{1}{2} m_{s_i}} C^{j_j m_j}_{l_j m_{l_i} \frac{1}{2} m_{s_i}} \int_{r'} g^*_{n_i l_j}(r') u_{n_j l_j \tau_j}(r') r'^2 dr' \delta_{m_{\tau_i} m_{\tau_j}} \phi_{n_i l_j j_i m_i}(\boldsymbol{r}) \tag{1-10}$$

$$= \sum_{n_i j_i m_i} \delta_{j_i j_j} \delta_{m_i m_j} \int_{r'} g^*_{n_i l_j}(r') u_{n_j l_j \tau_j}(r') r'^2 dr' \delta_{m_{\tau_i} m_{\tau_j}} \phi_{n_i l_j j_i m_i}(\boldsymbol{r}) \tag{1-11}$$

$$= \sum_{n_i} \int_{r'} g^*_{n_i l_j}(r') u_{n_j l_j \tau_j}(r') r'^2 dr' \delta_{m_{\tau_i} m_{\tau_j}} \phi_{n_i l_j j_j m_j}(\boldsymbol{r}) \tag{1-12}$$

For any one-body matrix element, we can use the following method to derive and calculate it.

$$\langle \bar{a} \| \mathcal{M}^{\lambda}_{1b} \| \bar{b} \rangle = \left\langle n_a l_a j_a \tau_a \middle\| \mathcal{M}^{\lambda}_{1b} \middle\| n_b l_b j_b \tau_b \right\rangle \tag{1-13}$$

$$= \frac{\hat{j}_a}{C^{j_a m_a}_{j_b m_b 1\mu}} \left\langle n_a l_a j_a m_a \tau_a \middle| \mathcal{M}^{\lambda\mu}_{1b} \middle| n_b l_b j_b m_b \tau_b \right\rangle \tag{1-14}$$

$$= \sum_{n_1 n_2} c^{n_a}_{n_1} c^{n_b}_{n_2} \frac{\hat{j}_a}{C^{j_a m_a}_{j_b m_b 1\mu}} \left\langle n_1 l_a j_a m_a \tau_a \middle| \mathcal{M}^{\lambda\mu}_{1b} \middle| n_2 l_b j_b m_b \tau_b \right\rangle \tag{1-15}$$

$$= \sum_{n_1 n_2} c^{n_a}_{n_1} c^{n_b}_{n_2} \left\langle n_1 l_a j_a \tau_a \middle\| \mathcal{M}^{\lambda}_{1b} \middle\| n_2 l_b j_b \tau_b \right\rangle \tag{1-16}$$

For any two-body matrix element, we can use the following method to derive and calculate it.

$$\left\langle \bar{a}\bar{a}' \middle\|^{J_{aa'}} \mathcal{M}^{\lambda}_{2b} \middle\| \bar{b}\bar{b}' \right\rangle^{J_{bb'}} = \left\langle n_a l_a j_a \tau_a, n_{a'} l_{a'} j_{a'} \tau_{a'} \middle\|^{J_{aa'}} \mathcal{M}^{\lambda}_{2b} \middle\| n_b l_b j_b \tau_b, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{bb'}}$$

$$= \frac{\hat{J}_{aa'}}{C^{J_{aa'} M_{aa'}}_{J_{bb'} M_{bb'} \lambda\mu}} \left\langle n_a l_a j_a \tau_a, n_{a'} l_{a'} j_{a'} \tau_{a'} \middle|^{J_{aa'} M_{aa'}} \mathcal{M}^{\lambda\mu}_{2b} \middle| n_b l_b j_b \tau_b, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{bb'} M_{bb'}} \tag{1-17}$$

$$= \frac{\hat{J}_{aa'}}{C^{J_{aa'} M_{aa'}}_{J_{bb'} M_{bb'} \lambda\mu}} \sum_{m_a m_{a'} m_b m_{b'}} C^{J_{aa'} M_{aa'}}_{j_a m_a j_{a'} m_{a'}} C^{J_{bb'} M_{bb'}}_{j_b m_b j_{b'} m_{b'}} \left\langle n_a l_a j_a m_a \tau_a, n_{a'} l_{a'} j_{a'} m_{a'} \tau_{a'} \middle| \mathcal{M}^{\lambda\mu}_{2b} \middle| n_b l_b j_b m_b \tau_b, n_{b'} l_{b'} j_{b'} m_{b'} \tau_{b'} \right\rangle \tag{1-18}$$

$$= \sum_{n_1 n_2 n_3 n_4} c^{n_a}_{n_1} c^{n_{a'}}_{n_2} c^{n_b}_{n_3} c^{n_{b'}}_{n_4} \frac{\hat{J}_{aa'}}{C^{J_{aa'} M_{aa'}}_{J_{bb'} M_{bb'} \lambda\mu}} \sum_{m_a m_{a'} m_b m_{b'}} C^{J_{aa'} M_{aa'}}_{j_a m_a j_{a'} m_{a'}} C^{J_{bb'} M_{bb'}}_{j_b m_b j_{b'} m_{b'}}$$
$$\cdot \left\langle n_1 l_a j_a m_a \tau_a, n_2 l_{a'} j_{a'} m_{a'} \tau_{a'} \middle| \mathcal{M}^{\lambda\mu}_{2b} \middle| n_3 l_b j_b m_b \tau_b, n_4 l_{b'} j_{b'} m_{b'} \tau_{b'} \right\rangle \tag{1-19}$$

$$= \sum_{n_1 n_2 n_3 n_4} c^{n_a}_{n_1} c^{n_{a'}}_{n_2} c^{n_b}_{n_3} c^{n_{b'}}_{n_4} \left\langle n_1 l_a j_a \tau_a, n_2 l_{a'} j_{a'} \tau_{a'} \middle\|^{J_{aa'}} \mathcal{M}^{\lambda}_{2b} \middle\| n_3 l_b j_b \tau_b, n_4 l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{bb'}} \tag{1-20}$$

Chapter2 Derivation of Reduced Matrix Element of Gamow-Teller Operator

The form of the atomic transition matrix element in beta decay[9],

$$\langle F|H_\beta|I\rangle = -\frac{G_\beta}{\sqrt{2}}l^\mu \int d^3x \langle f|e^{-i\boldsymbol{q}\cdot\boldsymbol{x}}\mathbb{J}_\mu(\boldsymbol{x})|i\rangle. \tag{2-1}$$

The form of the nuclear transition matrix element is

$$\langle 1_\nu^+|\mathbb{M}^{\lambda\mu}|0^+\rangle = \int d^3x \langle 1_\nu^+|e^{-i\boldsymbol{q}\cdot\boldsymbol{x}}\mathbb{J}_\mu(\boldsymbol{x})|0^+\rangle = \int d^3x \langle 1_\nu^+|\mathbb{J}_\mu(\boldsymbol{x})|0^+\rangle. \tag{2-2}$$

where,

$$\mathbb{J}_\mu(\boldsymbol{x}) = \mathbb{J}_{1b}(\boldsymbol{x}) + \mathbb{J}_{2b}(\boldsymbol{x}) \tag{2-3}$$

The following reduced matrix elements are all derived in the spherical harmonic oscillator basis.

2.1 The reduced matrix element of the GT one-body current operator

The one-body current operator is

$$\mathbb{J}_{1b}(\boldsymbol{x}) = -g_A\sum_i^A \boldsymbol{\sigma}_i\boldsymbol{\tau}_i^\pm \delta(\boldsymbol{x}-\boldsymbol{r}_k). \tag{2-4}$$

The resulting reduced matrix element is given by:

$$\left\langle n_a l_a j_a \tau_a \middle\| g_A\boldsymbol{\sigma}\boldsymbol{\tau}^\pm \middle\| n_b l_b j_b \tau_b \right\rangle \tag{2-5}$$

$$= g_A\delta_{\tau_a,\pm\frac{1}{2}}\delta_{\tau_b,\mp\frac{1}{2}}\left\langle n_a l_a j_a \tau_a \middle| n_b l_b j_b \tau_b \right\rangle \left\langle l_a \frac{1}{2} j_a \middle\| \boldsymbol{\sigma} \middle\| l_b \frac{1}{2} j_b \right\rangle \tag{2-6}$$

$$\overset{(A-17)}{=\!=\!=} g_A\left\langle n_a l_a j_a \tau_a \middle| n_b l_b j_b \tau_b \right\rangle \delta_{\tau_a,\pm\frac{1}{2}}\delta_{\tau_b,\mp\frac{1}{2}}\delta_{l_a l_b}(-)^{1+l_b+\frac{1}{2}+j_a}\hat{j}_a\hat{j}_b \begin{Bmatrix} j_b & j_a & 1 \\ \frac{1}{2} & \frac{1}{2} & l_b \end{Bmatrix} \left\langle \frac{1}{2} \middle\| \boldsymbol{\sigma} \middle\| \frac{1}{2} \right\rangle \tag{2-7}$$

$$\overset{(A-3)}{=\!=\!=} g_A\left\langle n_a l_a j_a \tau_a \middle| n_b l_b j_b \tau_b \right\rangle \delta_{\tau_a,\pm\frac{1}{2}}\delta_{\tau_b,\mp\frac{1}{2}}\delta_{l_a l_b}(-)^{l_b+\frac{3}{2}+j_a}\sqrt{6}\hat{j}_a\hat{j}_b \begin{Bmatrix} j_b & j_a & 1 \\ \frac{1}{2} & \frac{1}{2} & l_b \end{Bmatrix} \tag{2-8}$$

2.2 The reduced matrix element of the GT two-body currents operator

The two-body currents operator is[11]

$$\mathbb{J}_{2b}(\boldsymbol{x}) = \sum_{k<l}^A \boldsymbol{J}_{kl}(\boldsymbol{x}) = \frac{1}{2}\sum_{k\neq l}^A \boldsymbol{J}_{kl}(\boldsymbol{x}) \tag{2-9}$$

$$\boldsymbol{J}_{kl}(x) = \frac{2c_3 g_A}{m_N F_\pi^2}\left\{m_\pi^2\left[\left(\frac{\boldsymbol{\sigma}_l}{3} - \boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\right)Y_2(r) - \frac{\boldsymbol{\sigma}_l}{3}Y_0(r)\right] + \frac{\boldsymbol{\sigma}_l}{3}\delta(\boldsymbol{r})\right\}\boldsymbol{\tau}_l^{\pm}\delta(\boldsymbol{x}-\boldsymbol{r}_k) + (k\leftrightarrow l)$$
$$+\left(c_4+\frac{1}{4}\right)\frac{g_A}{2m_N F_\pi^2}\left\{m_\pi^2\left[\left(\frac{\boldsymbol{\sigma}_\times}{3} - \boldsymbol{\sigma}_k\times\hat{\boldsymbol{r}}\,\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\right)Y_2(r) - \frac{\boldsymbol{\sigma}_\times}{3}Y_0(r)\right] + \frac{\boldsymbol{\sigma}_\times}{3}\delta(\boldsymbol{r})\right\}\boldsymbol{\tau}_\times^{\pm}\delta(\boldsymbol{x}-\boldsymbol{r}_k) + (k\leftrightarrow l)$$
$$+\frac{ig_A}{8m_N F_\pi^2}\boldsymbol{\tau}_\times^{\pm}\left[\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\,m_\pi^2\left(1+\frac{2}{m_\pi r}+\frac{2}{m_\pi^2 r^2}\right)Y_0(r) + \frac{\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}}{r^2}Y_1(r) - \frac{\boldsymbol{\sigma}_l}{r^2}Y_1(r)\right]\delta(\boldsymbol{x}-\boldsymbol{r}_k) + (k\leftrightarrow l)$$
$$-\frac{g_A}{4m_N F_\pi^2}\left[2d_1(\boldsymbol{\sigma}_k\boldsymbol{\tau}_k^{\pm} + \boldsymbol{\sigma}_l\boldsymbol{\tau}_l^{\pm}) + d_2\boldsymbol{\sigma}_\times\boldsymbol{\tau}_\times^{\pm}\right]\delta(\boldsymbol{r})\delta(\boldsymbol{x}-\boldsymbol{r}_k) \tag{2-10}$$

where,

$$c_i = m_N\bar{c}_i\ , \qquad d_i = \frac{m_N F_\pi^2}{g_A}\bar{d}_i \tag{2-11}$$

$$m_\pi = 139.57\ MeV, \qquad F_\pi = 92.4\ MeV \tag{2-12}$$

$$\boldsymbol{r} = (\boldsymbol{r}_k - \boldsymbol{r}_l),\ \boldsymbol{\sigma}_\times = \boldsymbol{\sigma}_k\times\boldsymbol{\sigma}_l,\ \boldsymbol{\tau}_\times^{\pm} = (\boldsymbol{\tau}_k\times\boldsymbol{\tau}_l)^{\pm} \tag{2-13}$$

$$Y_0(r) = \frac{e^{-m_\pi r}}{4\pi r} \qquad [fm^{-1}] \tag{2-14}$$

$$Y_1(r) = -r\frac{\partial}{\partial r}Y_0(r) = (m_\pi r+1)Y_0(r) \qquad [fm^{-1}] \tag{2-15}$$

$$Y_2(r) = \frac{1}{m_\pi^2}r\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial}{\partial r}Y_0(r) = \left(1+\frac{3}{m_\pi r}+\frac{3}{m_\pi^2 r^2}\right)Y_0(r) \qquad [fm^{-1}] \tag{2-16}$$

Term$[01A_1]$: (Note: The variable r in $Y_2(r)$ and $Y_2(\sqrt{2}r)$ represents distinct quantities and please distinguish between them. The same applies to all other terms.)

$$[01A_1] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_l\, Y_2(r)\boldsymbol{\tau}_l^{-} \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-17}$$

$$= \left\langle n_a l_a j_a, n_b l_b j_b \right\|^{J_{ab}} [Y_2(\sqrt{2}r)\otimes\boldsymbol{\sigma}_l]^1 \left\| n_{a'} l_{a'} j_{a'}, n_{b'} l_{b'} j_{b'} \right\rangle^{J_{a'b'}} \langle \tau_a\tau_b|\boldsymbol{\tau}_l^{-}|\tau_{a'}\tau_{b'}\rangle \tag{2-18}$$

$$\overset{(A-4)}{=\!=} \langle \tau_a\tau_b|\boldsymbol{\tau}_l^{-}|\tau_{a'}\tau_{b'}\rangle \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \left\langle [n_a l_a\otimes n_b l_b]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \right\|^{J_{ab}} [Y_2(\sqrt{2}r)\otimes\boldsymbol{\sigma}_l]^1 \left\| [n_{a'} l_{a'}\otimes n_{b'} l_{b'}]^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle^{J_{a'b'}} \tag{2-19}$$

$$\underset{(A-6)}{\overset{(A-5)}{=\!=}} \langle \tau_a\tau_b|\boldsymbol{\tau}_l^{-}|\tau_{a'}\tau_{b'}\rangle \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'N'L'} \langle\langle NL, nl|n_a l_a, n_b l_b\rangle\rangle^{L_{ab}} \langle\langle N'L', n'l'|n_{a'} l_{a'}, n_{b'} l_{b'}\rangle\rangle^{L_{a'b'}}$$
$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$

$$\cdot \left\langle [NL \otimes nl]^{L_{ab}}[\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \right\|^{J_{ab}} \left[Y_2(\sqrt{2}r) \otimes \boldsymbol{\sigma}_l\right]^1 \left\| [N'L' \otimes n'l']^{L_{a'b'}}[\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \right\rangle^{J_{a'b'}} \tag{2-20}$$

$$\overset{(A-10)}{\underset{(A-1)}{=\!=\!=}} \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$

$$\cdot \sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'} \begin{Bmatrix} 0 & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \left\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \middle\| \boldsymbol{\sigma}_l \middle\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \right\rangle$$

$$\cdot \left\langle [NL \otimes nl]^{L_{ab}} \middle\| Y_2(\sqrt{2}r) \middle\| [N'L' \otimes n'l']^{L_{a'b'}} \right\rangle \tag{2-21}$$

$$\overset{[5]}{\underset{P357(2)}{=\!=\!=}} \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$

$$\cdot \sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\delta_{L_{ab}L_{a'b'}}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}} \frac{1}{\sqrt{3}\hat{L}_{ab}} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}$$

$$\cdot \left\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \middle\| \boldsymbol{\sigma}_l \middle\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \right\rangle \left\langle [NL \otimes nl]^{L_{ab}} \middle\| Y_2(\sqrt{2}r) \middle\| [N'L' \otimes n'l']^{L_{a'b'}} \right\rangle \tag{2-22}$$

$$\overset{(B-11)}{\underset{(B-2)}{=\!=\!=}} \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$

$$\cdot \sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\delta_{L_{ab}L_{a'b'}}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}} \frac{1}{\sqrt{3}\hat{L}_{ab}} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}$$

$$\cdot (-)^{S_{ab}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \delta_{ll'}\delta_{NN'}\delta_{LL'}\delta_{L_{ab}L_{a'b'}}\hat{L}_{ab} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-23}$$

$$= \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}}\sqrt{6}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{ab}+S_{a'b'}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{S}^2_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle NL,n'l|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{ab}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle \underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-24}$$

In the code:(101)

$$[01A_1] = \frac{2\sqrt{6}c_3g_Am_\pi^2}{3m_NF_\pi^2}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{ab}+S_{a'b'}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{S}^2_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle NL,n'l|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{ab}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle \underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-25}$$

Term$[02B_1]$:

$$[02B_1] = \Big\langle n_al_aj_a\tau_a, n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}Y_2(r)\boldsymbol{\tau}_l^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'}, n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-26}$$
$$\xlongequal[(A-6)]{(A-4)(A-5)}\langle\tau_a\tau_b|\boldsymbol{\tau}_l^-|\tau_{a'}\tau_{b'}\rangle\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'l'N'L'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle N'L',n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$
$$\cdot\Big\langle[NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\Big\|^{J_{ab}}\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}Y_2(\sqrt{2}r)\Big\|[N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\Big\rangle^{J_{a'b'}} \tag{2-27}$$

$$\xlongequal[(B-15)]{(A-1)}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'l'N'L'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle N'L',n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}$$

$$\cdot \left\langle [NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\right\|^{J_{ab}}\left[\boldsymbol{\sigma}_l\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\right]^1 Y_2(\sqrt{2}r)\left\|[N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle^{J_{a'b'}} \tag{2-28}$$

$$\overset{(A-11)}{=\!=\!=} \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle N'L',n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}$$

$$\cdot (-)^h\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\begin{Bmatrix} h & 1 & 1\\ L_{ab} & S_{ab} & J_{ab}\\ L_{a'b'} & S_{a'b'} & J_{a'b'}\end{Bmatrix}\left\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\right\|\boldsymbol{\sigma}_l\left\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle$$

$$\cdot \left\langle [NL\otimes nl]^{L_{ab}}\right\|Y_2(\sqrt{2}r)[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\left\|[N'L'\otimes n'l']^{L_{a'b'}}\right\rangle \tag{2-29}$$

$$\overset{(B-24)}{\underset{(B-2)}{=\!=\!=}} \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle N'L',n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}$$

$$\cdot (-)^h\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\begin{Bmatrix} h & 1 & 1\\ L_{ab} & S_{ab} & J_{ab}\\ L_{a'b'} & S_{a'b'} & J_{a'b'}\end{Bmatrix}(-)^{S_{ab}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}$$

$$\cdot\, \delta_{NN'}\delta_{LL'}(-)^{h+L+l'+L_{ab}}C_{1010}^{h0}C_{l'0h0}^{l0}\hat{l}'\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{ab} & L_{a'b'} & h\\ l' & l & L'\end{Bmatrix}\left\langle\underbrace{nl}_{r}\middle|Y_2(\sqrt{2}r)\middle|n'l'\right\rangle \tag{2-30}$$

$$= \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^1\sqrt{6}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}+S_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2\begin{Bmatrix} S_{a'b'} & S_{ab} & 1\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h\hat{h}\begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010}C^{l0}_{l'0h0}\begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-31}$$

In the code：(102)

$$[02B_1] = \frac{2\sqrt{6}c_3g_Am_\pi^2}{m_NF_\pi^2}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}+S_{ab}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h\hat{h}\begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010}C^{l0}_{l'0h0}\begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-32}$$

Term$[03C_1]$：(The derivation follows the same method as Term$[01A_1]$)

$$[03C_1] = \Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_lY_0(r)\boldsymbol{\tau}_l^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-33}$$

$$= \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}}\sqrt{6}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{ab}+S_{a'b'}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{S}^2_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle \underbrace{nl}_{r}\Big|Y_0(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-34}$$

In the code：(103)

$$[03C_1] = \frac{2\sqrt{6}c_3g_Am_\pi^2}{3m_NF_\pi^2}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{J_{ab}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}+S_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| Y_0(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-35}$$

Term$[04D_1]$:

$$[04D_1] = \Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \Big\|^{J_{ab}} \boldsymbol{\sigma}_l \, \delta(\boldsymbol{r}) \boldsymbol{\tau}_l^- \Big\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \Big\rangle^{J_{a'b'}} \tag{2-36}$$

$$\xlongequal{(A-4)} \langle \tau_a \tau_b | \boldsymbol{\tau}_l^- | \tau_{a'} \tau_{b'} \rangle \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\|^{J_{ab}} [\delta(\boldsymbol{r}) \otimes \boldsymbol{\sigma}_l]^1 \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle^{J_{a'b'}} \tag{2-37}$$

$$\xlongequal[(A-1)]{(A-10)} \delta_{\tau_a \tau_{a'}} \delta_{\tau_b, -\frac{1}{2}} \delta_{\tau_{b'}, \frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sqrt{3} \hat{J}_{ab} \hat{J}_{a'b'} \begin{Bmatrix} 0 & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} \Big\| \delta(\boldsymbol{r}) \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} \Big\rangle$$
$$\cdot \Big\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\| \boldsymbol{\sigma}_l \Big\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle \tag{2-38}$$

$$\xlongequal[P357(2)]{[5]} \delta_{\tau_a \tau_{a'}} \delta_{\tau_b, -\frac{1}{2}} \delta_{\tau_{b'}, \frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sqrt{3} \hat{J}_{ab} \hat{J}_{a'b'} (-)^{1+L_{ab}+S_{a'b'}+J_{ab}} \frac{1}{\sqrt{3} \hat{L}_{ab}} \delta_{L_{ab} L_{a'b'}} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}$$
$$\cdot \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} \Big\| \delta(\boldsymbol{r}) \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} \Big\rangle \Big\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\| \boldsymbol{\sigma}_l \Big\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle \tag{2-39}$$

$$\xlongequal[(B-36)]{(B-2)} \delta_{\tau_a \tau_{a'}} \delta_{\tau_b, -\frac{1}{2}} \delta_{\tau_{b'}, \frac{1}{2}} \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}}\frac{1}{\sqrt{3}\hat{L}_{ab}}\delta_{L_{ab}L_{a'b'}}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}(-)^{S_{ab}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}$$

$$\cdot\sum_{l=0}^{\infty}(-)^{l_{a'}+l_b+L_{ab}}\frac{1}{4\pi}\delta_{L_{ab}L_{a'b'}}\,\hat{l}^2\,\hat{l}_{a'}\hat{l}_{b'}\hat{L}_{ab}\,C^{l_a0}_{l_{a'}0l0}\,C^{l_b0}_{l_{b'}0l0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l\end{Bmatrix}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle \tag{2-40}$$

$$=\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}+l_{a'}+l_b}\frac{\sqrt{6}}{4\pi}\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$

$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{S_{a'b'}+S_{ab}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{S}^2_{a'b'}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C^{l_a0}_{l_{a'}0l0}\,C^{l_b0}_{l_{b'}0l0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l\end{Bmatrix} \tag{2-41}$$

In the code：(104)

$$[04D_1]=\frac{c_3g_A}{\sqrt{6}\pi m_NF_\pi^2}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}+l_{a'}+l_b}\,\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$

$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{S_{a'b'}+S_{ab}}\hat{L}^2_{ab}\hat{S}^2_{ab}\hat{S}^2_{a'b'}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C^{l_a0}_{l_{a'}0l0}\,C^{l_b0}_{l_{b'}0l0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l\end{Bmatrix} \tag{2-42}$$

Term$[05E_1]$:

$$[05E_1]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_\times Y_2(r)\boldsymbol{\tau}^-_\times\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-43}$$

$$=\Big\langle n_al_aj_a,n_bl_bj_b\Big\|^{J_{ab}}[Y_2(\sqrt{2}r)\otimes\boldsymbol{\sigma}_\times]^1\Big\|n_{a'}l_{a'}j_{a'},n_{b'}l_{b'}j_{b'}\Big\rangle^{J_{a'b'}}\langle\tau_a\tau_b|\boldsymbol{\tau}^-_\times|\tau_{a'}\tau_{b'}\rangle \tag{2-44}$$

$$\stackrel{(A-4)(A-5)}{\overline{\overline{\;(A-6)\;}}}\langle\tau_a\tau_b|\boldsymbol{\tau}^-_\times|\tau_{a'}\tau_{b'}\rangle\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$

$$\cdot\sum_{nlNL,n'l'N'L'}\langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}}\langle\langle N'L',n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$

$$\cdot\Big\langle[NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\Big\|^{J_{ab}}[Y_2(\sqrt{2}r)\otimes\boldsymbol{\sigma}_\times]^1\Big\|[N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\Big\rangle^{J_{a'b'}} \tag{2-45}$$

$$
\xlongequal[(B-43)P357(2)]{(A-10)[5]} 4i\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)
$$

$$
\cdot\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}
$$

$$
\cdot\sum_{nlNL,n'l'N'L'}\big\langle\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\big\rangle^{L_{ab}}\big\langle\big\langle N'L',n'l'\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\big\rangle^{L_{a'b'}}
$$

$$
\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}
$$

$$
\cdot\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\delta_{L_{ab}L_{a'b'}}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}}\frac{1}{\sqrt{3}\hat{L}_{ab}}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}
$$

$$
\cdot\Big\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\Big\|\boldsymbol{\sigma}_\times\Big\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\Big\rangle\Big\langle[NL\otimes nl]^{L_{ab}}\Big\|Y_2(\sqrt{2}r)\Big\|[N'L'\otimes n'l']^{L_{a'b'}}\Big\rangle \tag{2-46}
$$

$$
\xlongequal[(B-53)]{(B-11)} 4i\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)
$$

$$
\cdot\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}
$$

$$
\cdot\sum_{nlNL,n'l'N'L'}\big\langle\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\big\rangle^{L_{ab}}\big\langle\big\langle N'L',n'l'\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\big\rangle^{L_{a'b'}}
$$

$$
\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}
$$

$$
\cdot\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}\delta_{L_{ab}L_{a'b'}}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}}\frac{1}{\sqrt{3}\hat{L}_{ab}}\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}
$$

$$
\cdot(-6\sqrt{6}i)\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix} 1 & 1 & 1\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\end{Bmatrix}\delta_{ll'}\delta_{NN'}\delta_{LL'}\delta_{L_{ab}L_{a'b'}}\hat{L}_{ab}\Big\langle\underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-47}
$$

$$
=(-)^{1+J_{ab}}\,24\sqrt{6}\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)
$$

$$
\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}\begin{Bmatrix} 1 & 1 & 1\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\end{Bmatrix}
$$

$$
\cdot\sum_{nlNL,n'}\big\langle\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\big\rangle^{L_{ab}}\big\langle\big\langle NL,n'l\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\big\rangle^{L_{ab}}
$$

$$
\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle\underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-48}
$$

In the code：(105)

$$
\begin{aligned}
[05E_1] =& \Big(c_4+\frac{1}{4}\Big)\frac{4\sqrt{6}g_A m_\pi^2}{m_N F_\pi^2}(-)^{1+J_{ab}}\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)\\
&\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}\begin{Bmatrix} 1 & 1 & 1\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\end{Bmatrix}\\
&\cdot\sum_{nlNL,n'}\big\langle\!\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\!\big\rangle^{L_{ab}}\big\langle\!\big\langle NL,n'l\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\!\big\rangle^{L_{ab}}\\
&\cdot\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle\underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l\Big\rangle
\end{aligned}
\tag{2-49}
$$

Term$[06F_1]$：

$$
[06F_1]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_k\times\hat{\boldsymbol{r}}\,\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\,Y_2(r)\boldsymbol{\tau}_\times^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-50}
$$

$$
\begin{aligned}
\overset{(A-4)(A-5)}{\underset{(A-6)}{=\!=}}&\big\langle\tau_a\tau_b\big|\boldsymbol{\tau}_\times^-\big|\tau_{a'}\tau_{b'}\big\rangle\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}\\
&\cdot\sum_{nlNL,n'l'N'L'}\big\langle\!\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\!\big\rangle^{L_{ab}}\big\langle\!\big\langle N'L',n'l'\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\!\big\rangle^{L_{a'b'}}\\
&\cdot\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\\
&\cdot\Big\langle[NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\Big\|^{J_{ab}}\boldsymbol{\sigma}_k\times\hat{\boldsymbol{r}}\,\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\,Y_2(\sqrt{2}r)\Big\|[N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\Big\rangle^{J_{a'b'}}
\end{aligned}
\tag{2-51}
$$

$$
\begin{aligned}
\overset{(B-58)}{\underset{(B-43)}{=\!=}}&4i\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)\\
&\cdot\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}\\
&\cdot\sum_{nlNL,n'l'N'L'}\big\langle\!\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\!\big\rangle^{L_{ab}}\big\langle\!\big\langle N'L',n'l'\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\!\big\rangle^{L_{a'b'}}\\
&\cdot\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\sqrt{2}i\sum_{g,h=0}^{2}\hat{g}\,\hat{h}\,(-)^{h+1}\begin{Bmatrix} 1 & 1 & 1\\ g & h & 1\end{Bmatrix}\\
&\cdot\Big\langle[NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\Big\|^{J_{ab}}\big[[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^g\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\big]^1\,Y_2(\sqrt{2}r)\Big\|[N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\Big\rangle^{J_{a'b'}}
\end{aligned}
\tag{2-52}
$$

$$
\overset{(A-11)}{=\!=}4i\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)
$$

$$
\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}
$$

$$
\cdot \sum_{nlNL,n'l'N'L'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle N'L', n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}
$$

$$
\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \sqrt{2}i \sum_{g,h=0}^{2} \hat{g}\,\hat{h}\,(-)^{h+1} \begin{Bmatrix} 1 & 1 & 1 \\ g & h & 1 \end{Bmatrix}
$$

$$
\cdot (-)^{g+h+1}\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'} \begin{Bmatrix} h & g & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \left\langle [\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \middle\| [\boldsymbol{\sigma}_k \otimes \boldsymbol{\sigma}_l]^g \middle\| [\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle
$$

$$
\cdot \left\langle [NL\otimes nl]^{L_{ab}} \middle\| Y_2(\sqrt{2}r)[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h \middle\| [N'L'\otimes n'l']^{L_{a'b'}} \right\rangle \tag{2-53}
$$

$$
\overset{(B-61)}{\underset{(B-24)}{=\!=}} 4i\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)
$$

$$
\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}
$$

$$
\cdot \sum_{nlNL,n'l'N'L'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle N'L', n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}
$$

$$
\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \sqrt{2}i \sum_{g,h=0}^{2} \hat{g}\,\hat{h}\,(-)^{h+1} \begin{Bmatrix} 1 & 1 & 1 \\ g & h & 1 \end{Bmatrix}
$$

$$
\cdot (-)^{g+h+1}\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'} \begin{Bmatrix} h & g & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} 6\hat{g}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} 1 & 1 & g \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}
$$

$$
\cdot \delta_{NN'}\delta_{LL'}(-)^{h+l'+L'+L_{ab}} C^{h0}_{1010} C^{l0}_{l'0h0} \hat{l}' \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle \tag{2-54}
$$

$$
= (-)^1 24\sqrt{6}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)
$$

$$
\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}} \hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}
$$

$$
\cdot \sum_{nlNL,n'l'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}
$$

$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle\underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot\sum_{g,h=0}^{2}(-)^{g+h}\hat{g}^2\,\hat{h}\,C_{1010}^{h0}C_{l'0h0}^{l0}\begin{Bmatrix}1&1&1\\g&h&1\end{Bmatrix}\begin{Bmatrix}L_{ab}&L_{a'b'}&h\\l'&l&L\end{Bmatrix}\begin{Bmatrix}h&g&1\\L_{ab}&S_{ab}&J_{ab}\\L_{a'b'}&S_{a'b'}&J_{a'b'}\end{Bmatrix}\begin{Bmatrix}1&1&g\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix}\tag{2-55}$$

In the code：(106)

$$[06F_1]=\left(c_4+\frac{1}{4}\right)\frac{12\sqrt{6}g_Am_\pi^2}{m_NF_\pi^2}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2\begin{Bmatrix}l_a&l_b&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\j_a&j_b&J_{ab}\end{Bmatrix}\begin{Bmatrix}l_{a'}&l_{b'}&L_{a'b'}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\\j_{a'}&j_{b'}&J_{a'b'}\end{Bmatrix}$$

$$\cdot\sum_{nlNL,n'l'}\big\langle\!\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\!\big\rangle^{L_{ab}}\big\langle\!\big\langle NL,n'l'\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\!\big\rangle^{L_{a'b'}}$$

$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle\underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot\sum_{g,h=0}^{2}(-)^{g+h}\hat{g}^2\,\hat{h}\,C_{1010}^{h0}C_{l'0h0}^{l0}\begin{Bmatrix}1&1&1\\g&h&1\end{Bmatrix}\begin{Bmatrix}L_{ab}&L_{a'b'}&h\\l'&l&L\end{Bmatrix}\begin{Bmatrix}h&g&1\\L_{ab}&S_{ab}&J_{ab}\\L_{a'b'}&S_{a'b'}&J_{a'b'}\end{Bmatrix}\begin{Bmatrix}1&1&g\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix}\tag{2-56}$$

Term$[07G_1]$：(The derivation follows the same method as Term$[05E_1]$)

$$[07G_1]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_\times Y_0(r)\boldsymbol{\tau}_\times^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}}\tag{2-57}$$

$$=(-)^{1+J_{ab}}24\sqrt{6}\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix}J_{a'b'}&J_{ab}&1\\S_{ab}&S_{a'b'}&L_{ab}\end{Bmatrix}\begin{Bmatrix}l_a&l_b&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\j_a&j_b&J_{ab}\end{Bmatrix}\begin{Bmatrix}l_{a'}&l_{b'}&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\\j_{a'}&j_{b'}&J_{a'b'}\end{Bmatrix}\begin{Bmatrix}1&1&1\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix}$$

$$\cdot\sum_{nlNL,n'}\big\langle\!\big\langle NL,nl\big|n_al_a,n_bl_b\big\rangle\!\big\rangle^{L_{ab}}\big\langle\!\big\langle NL,n'l\big|n_{a'}l_{a'},n_{b'}l_{b'}\big\rangle\!\big\rangle^{L_{ab}}$$

$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle\underbrace{nl}_{r}\Big|Y_0(\sqrt{2}r)\Big|n'l\Big\rangle\tag{2-58}$$

In the code：(107)

$$[07G_1]=\left(c_4+\frac{1}{4}\right)\frac{4\sqrt{6}g_Am_\pi^2}{m_NF_\pi^2}(-)^{J_{ab}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl \big| n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l \big| n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| Y_0(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-59}$$

Term$[08H_1]$:

$$[08H_1] = \Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \Big\| ^{J_{ab}} \boldsymbol{\sigma}_\times \delta(\boldsymbol{r}) \boldsymbol{\tau}_\times^- \Big\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \Big\rangle^{J_{a'b'}} \tag{2-60}$$

$$\xlongequal{(A-4)} \big\langle \tau_a \tau_b \big| \boldsymbol{\tau}_\times^- \big| \tau_{a'} \tau_{b'} \big\rangle \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\| ^{J_{ab}} [\delta(\boldsymbol{r}) \otimes \boldsymbol{\sigma}_\times]^1 \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle^{J_{a'b'}} \tag{2-61}$$

$$\xlongequal[(B-43)]{(A-10)} 4i\Big(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a\Big)$$
$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sqrt{3} \hat{J}_{ab} \hat{J}_{a'b'} \begin{Bmatrix} 0 & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} \Big\| \delta(\boldsymbol{r}) \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} \Big\rangle$$
$$\cdot \Big\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\| \boldsymbol{\sigma}_\times \Big\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle \tag{2-62}$$

$$\xlongequal[P357(2)]{[5]} 4i\Big(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a\Big)$$
$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{L}_{ab} \hat{S}_{ab} \hat{L}_{a'b'} \hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sqrt{3} \hat{J}_{ab} \hat{J}_{a'b'} (-)^{1+L_{ab}+S_{a'b'}+J_{ab}} \frac{1}{\sqrt{3}\hat{L}_{ab}} \delta_{L_{ab}L_{a'b'}} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}$$
$$\cdot \Big\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} \Big\| \delta(\boldsymbol{r}) \Big\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} \Big\rangle \Big\langle [\frac{1}{2} \otimes \frac{1}{2}]^{S_{ab}} \Big\| \boldsymbol{\sigma}_\times \Big\| [\frac{1}{2} \otimes \frac{1}{2}]^{S_{a'b'}} \Big\rangle \tag{2-63}$$

$$\xlongequal[(B-53)]{(B-36)} 4i\Big(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a\Big)$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'}(-)^{1+L_{ab}+S_{a'b'}+J_{ab}} \frac{1}{\sqrt{3}\hat{L}_{ab}} \delta_{L_{ab}L_{a'b'}} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix}$$

$$\cdot \sum_{l=0}^{\infty}(-)^{l_{a'}+l_b+L_{ab}} \frac{1}{4\pi}\delta_{L_{ab}L_{a'b'}}\ \hat{l}^2\,\hat{l}_{a'}\hat{l}_{b'}\hat{L}_{ab}\, C_{l_{a'}0l0}^{l_a0}\, C_{l_{b'}0l0}^{l_b0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \Big\langle \underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big| n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$

$$\cdot (-6\sqrt{6})i\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \tag{2-64}$$

$$= \frac{6\sqrt{6}}{\pi}(-)^{1+l_{a'}+l_b+J_{ab}}\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \Big\langle \underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big| n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle \sum_{l=0}^{\infty} \hat{l}^2\, C_{l_{a'}0l0}^{l_a0}\, C_{l_{b'}0l0}^{l_b0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-65}$$

In the code：(108)

$$[08H_1] = \left(c_4+\frac{1}{4}\right)\frac{\sqrt{6}g_A}{\pi m_N F_\pi^2}(-)^{1+l_{a'}+l_b+J_{ab}}\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \Big\langle \underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big| n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle \sum_{l=0}^{\infty} \hat{l}^2\, C_{l_{a'}0l0}^{l_a0}\, C_{l_{b'}0l0}^{l_b0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-66}$$

Term$[09I_1]$：(The derivation follows the same methodology as Term$[04D_1]$)

$$[09I_1] = \Big\langle n_al_aj_a\tau_a, n_bl_bj_b\tau_b \Big\|^{J_{ab}} \boldsymbol{\sigma}_l\,\delta(\boldsymbol{r})\boldsymbol{\tau}_l^- \Big\| n_{a'}l_{a'}j_{a'}\tau_{a'}, n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-67}$$

$$= \delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{1+J_{ab}+l_{a'}+l_b}\frac{\sqrt{6}}{4\pi}\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle \underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big| n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}+S_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C_{l_{a'}0l0}^{l_a0}\,C_{l_{b'}0l0}^{l_b0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab}\\ l_a & l_b & l\end{Bmatrix} \tag{2-68}$$

In the code：(109)

$$[09I_1]=\frac{\sqrt{6}d_1g_A}{8\pi m_N F_\pi^2}\delta_{\tau_a\tau_{a'}}\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}(-)^{J_{ab}+l_{a'}+l_b}\,\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{S_{a'b'}+S_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C_{l_{a'}0l0}^{l_a0}\,C_{l_{b'}0l0}^{l_b0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab}\\ l_a & l_b & l\end{Bmatrix} \tag{2-69}$$

Term$[09I_2]$：(This term corresponds to the exchange term of $[09I_1](k\longleftrightarrow l)$)

$$[09I_2]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_k\,\delta(\boldsymbol{r})\boldsymbol{\tau}_k^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-70}$$

$$=\delta_{\tau_b\tau_{b'}}\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}(-)^{1+J_{ab}+l_{a'}+l_b}\frac{\sqrt{6}}{4\pi}\,\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C_{l_{a'}0l0}^{l_a0}\,C_{l_{b'}0l0}^{l_b0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab}\\ l_a & l_b & l\end{Bmatrix} \tag{2-71}$$

In the code：(110)

$$[09I_2]=\frac{\sqrt{6}d_1g_A}{8\pi m_N F_\pi^2}\delta_{\tau_b\tau_{b'}}\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}(-)^{J_{ab}+l_{a'}+l_b}\,\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\Big\langle\underbrace{n_al_a}_{r_1}\underbrace{n_bl_b}_{r_1}\Big|n_{a'}l_{a'}n_{b'}l_{b'}\Big\rangle$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix} J_{a'b'} & J_{ab} & 1\\ S_{ab} & S_{a'b'} & L_{ab}\end{Bmatrix}\begin{Bmatrix} S_{a'b'} & S_{ab} & 1\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{ab}\\ j_a & j_b & J_{ab}\end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{ab}\\ \frac{1}{2} & \frac{1}{2} & S_{a'b'}\\ j_{a'} & j_{b'} & J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{l=0}^{\infty}\hat{l}^2\,C_{l_{a'}0l0}^{l_a0}\,C_{l_{b'}0l0}^{l_b0}\begin{Bmatrix} l_{b'} & l_{a'} & L_{ab}\\ l_a & l_b & l\end{Bmatrix} \tag{2-72}$$

Term$[10J_1]$：(This term corresponds to the exchange term of $[08H_1]$)

$$[10J_1]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_\times\,\delta(\boldsymbol{r})\boldsymbol{\tau}_\times^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-73}$$

$$=\frac{6\sqrt{6}}{\pi}(-)^{1+l_{a'}+l_b+J_{ab}}\hat{l}_{a'}\hat{l}_{b'}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \left\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \middle| n_{a'} l_{a'} n_{b'} l_{b'} \right\rangle \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'}0l0}^{l_a 0} C_{l_{b'}0l0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-74}$$

In the code：(111)

$$[10J_1] = \frac{3\sqrt{6}d_2 g_A}{2\pi m_N F_\pi^2} (-)^{l_{a'}+l_b+J_{ab}} \hat{l}_{a'} \hat{l}_{b'} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \left\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \middle| n_{a'} l_{a'} n_{b'} l_{b'} \right\rangle \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'}0l0}^{l_a 0} C_{l_{b'}0l0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-75}$$

Term$[01A_2]$：(This term corresponds to the exchange term of $[01A_1](k \longleftrightarrow l)$)

$$[01A_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_k Y_2(r) \boldsymbol{\tau}_k^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-76}$$

$$= \delta_{\tau_b \tau_{b'}} \delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} (-)^{1+J_{ab}} \sqrt{6} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-77}$$

In the code：(112)

$$[01A_2] = \frac{2\sqrt{6}c_3 g_A m_\pi^2}{3 m_N F_\pi^2} \delta_{\tau_b \tau_{b'}} \delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} (-)^{1+J_{ab}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-78}$$

Term$[02B_2]$: (This term corresponds to the exchange term of $[02B_1](k \longleftrightarrow l)$)

$$[02B_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_k \cdot \hat{\boldsymbol{r}}\hat{\boldsymbol{r}} Y_2(r) \boldsymbol{\tau}_k^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-79}$$

$$= \delta_{\tau_b\tau_{b'}} \delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} (-)^1 \sqrt{6} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{L}_{a'b'}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle NL, n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} (-)^{l'+L} \hat{l}' \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle$$

$$\cdot \sum_{h=0}^{2} (-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0} C_{l'0h0}^{l0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-80}$$

In the code: (113)

$$[02B_2] = \frac{2\sqrt{6} c_3 g_A m_\pi^2}{m_N F_\pi^2} \delta_{\tau_b\tau_{b'}} \delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{L}_{a'b'}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle NL, n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} (-)^{l'+L} \hat{l}' \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle$$

$$\cdot \sum_{h=0}^{2} (-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0} C_{l'0h0}^{l0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-81}$$

Term$[03C_2]$: (This term corresponds to the exchange term of $[03C_1](k \longleftrightarrow l)$)

$$[03C_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_k Y_0(r) \boldsymbol{\tau}_k^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-82}$$

$$= \delta_{\tau_b\tau_{b'}} \delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} (-)^{1+J_{ab}} \sqrt{6} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| Y_0(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-83}$$

In the code：(114)

$$[03C_2] = \frac{2\sqrt{6} c_3 g_A m_\pi^2}{3 m_N F_\pi^2} \delta_{\tau_b \tau_{b'}} \delta_{\tau_a, -\frac{1}{2}} \delta_{\tau_{a'}, \frac{1}{2}} (-)^{J_{ab}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab} S_{ab} S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| Y_0(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-84}$$

Term$[04D_2]$：(This term corresponds to the exchange term of $[04D_1](k \longleftrightarrow l)$)

$$[04D_2] = \Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \Big\|^{J_{ab}} \boldsymbol{\sigma}_k \, \delta(\boldsymbol{r}) \boldsymbol{\tau}_k^- \Big\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \Big\rangle^{J_{a'b'}} \tag{2-85}$$

$$= \delta_{\tau_b \tau_{b'}} \delta_{\tau_a, -\frac{1}{2}} \delta_{\tau_{a'}, \frac{1}{2}} (-)^{1+J_{ab}+l_{a'}+l_b} \frac{\sqrt{6}}{4\pi} \hat{l}_{a'} \hat{l}_{b'} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle$$

$$\cdot \sum_{L_{ab} S_{ab} S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'} 0 l 0}^{l_a 0} C_{l_{b'} 0 l 0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-86}$$

In the code：(115)

$$[04D_2] = \frac{c_3 g_A}{\sqrt{6} \pi m_N F_\pi^2} \delta_{\tau_b \tau_{b'}} \delta_{\tau_a, -\frac{1}{2}} \delta_{\tau_{a'}, \frac{1}{2}} (-)^{1+J_{ab}+l_{a'}+l_b} \hat{l}_{a'} \hat{l}_{b'} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle$$

$$\cdot \sum_{L_{ab} S_{ab} S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'} 0 l 0}^{l_a 0} C_{l_{b'} 0 l 0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-87}$$

Term$[05E_2]$：(This term corresponds to the exchange term of $[05E_1](k \longleftrightarrow l)$，and$[05E_2] =$

$[05E_1]$)

$$[05E_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_\times Y_2(r) \boldsymbol{\tau}_\times^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-88}$$

$$= (-)^{1+J_{ab}} 24\sqrt{6}\, \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)$$
$$\cdot \sum_{L_{ab} S_{ab} S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-89}$$

In the code：(116)

$$[05E_2] = \left(c_4 + \frac{1}{4}\right) \frac{4\sqrt{6} g_A m_\pi^2}{m_N F_\pi^2} (-)^{1+J_{ab}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)$$
$$\cdot \sum_{L_{ab} S_{ab} S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-90}$$

Term$[06F_2]$：(This term corresponds to the exchange term of $[06F_1](k \longleftrightarrow l)$)

$$[06F_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_l \times \hat{\boldsymbol{r}}\, \boldsymbol{\sigma}_k \cdot \hat{\boldsymbol{r}}\, Y_2(r) \boldsymbol{\tau}_\times^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-91}$$

$$= (-)^1 24\sqrt{6} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)$$
$$\cdot \sum_{L_{ab} S_{ab} L_{a'b'} S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{L}_{a'b'}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}$$
$$\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l', 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} (-)^{l'+L} \hat{l}' \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle$$

$$\cdot \sum_{g,h=0}^{2} (-)^{g+1}(-)^{g+h}\hat{g}^2\,\hat{h}\,C_{1010}^{h0}C_{l'0h0}^{l0} \begin{Bmatrix} 1 & 1 & 1 \\ g & h & 1 \end{Bmatrix} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \begin{Bmatrix} h & g & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & g \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \tag{2-92}$$

In the code: (117)

$$[06F_2] = \left(c_4 + \frac{1}{4}\right)\frac{12\sqrt{6}g_A m_\pi^2}{m_N F_\pi^2}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2 \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|Y_2(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{g,h=0}^{2} (-)^{g+1}(-)^{g+h}\hat{g}^2\,\hat{h}\,C_{1010}^{h0}C_{l'0h0}^{l0} \begin{Bmatrix} 1 & 1 & 1 \\ g & h & 1 \end{Bmatrix} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \begin{Bmatrix} h & g & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & g \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \tag{2-93}$$

Term$[07G_2]$: (This term corresponds to the exchange term of $[07G_1](k \longleftrightarrow l)$, and $[07G_2] = [07G_1]$)

$$[07G_2] = \Big\langle n_al_aj_a\tau_a, n_bl_bj_b\tau_b\Big\|^{J_{ab}} \boldsymbol{\sigma}_\times\, Y_0(r)\boldsymbol{\tau}_\times^- \Big\| n_{a'}l_{a'}j_{a'}\tau_{a'}, n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}} \tag{2-94}$$

$$= (-)^{1+J_{ab}}\, 24\sqrt{6}\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'} \langle\langle NL,nl|n_al_a,n_bl_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l|n_{a'}l_{a'},n_{b'}l_{b'}\rangle\rangle^{L_{ab}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle \underbrace{nl}_{r}\Big|Y_0(\sqrt{2}r)\Big|n'l\Big\rangle \tag{2-95}$$

In the code: (118)

$$[07G_2] = \left(c_4 + \frac{1}{4}\right)\frac{4\sqrt{6}g_A m_\pi^2}{m_N F_\pi^2}(-)^{J_{ab}}\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$

$$
\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix}
$$
$$
\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}
$$
$$
\cdot \delta_{2N+L+2n+l, 2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l, 2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \left\langle \underbrace{nl}_{r} \middle| Y_0(\sqrt{2}r) \middle| n'l \right\rangle \tag{2-96}
$$

Term$[08H_2]$: (This term corresponds to the exchange term of $[08H_1](k \longleftrightarrow l)$, and$[08H_2] = [08H_1]$)

$$
[08H_2] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_\times \, \delta(\boldsymbol{r}) \boldsymbol{\tau}_\times^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}} \tag{2-97}
$$
$$
= \frac{6\sqrt{6}}{\pi} (-)^{1+l_{a'}+l_b+J_{ab}} \hat{l}_{a'} \hat{l}_{b'} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)
$$
$$
\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}
$$
$$
\cdot \left\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \middle| n_{a'} l_{a'} n_{b'} l_{b'} \right\rangle \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'}0l0}^{l_a 0} C_{l_{b'}0l0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-98}
$$

In the code: (119)

$$
[08H_2] = \left(c_4 + \frac{1}{4}\right) \frac{\sqrt{6} g_A}{\pi m_N F_\pi^2} (-)^{1+l_{a'}+l_b+J_{ab}} \hat{l}_{a'} \hat{l}_{b'} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'} \left(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a \right)
$$
$$
\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}
$$
$$
\cdot \left\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \middle| n_{a'} l_{a'} n_{b'} l_{b'} \right\rangle \sum_{l=0}^{\infty} \hat{l}^2 C_{l_{a'}0l0}^{l_a 0} C_{l_{b'}0l0}^{l_b 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \tag{2-99}
$$

The following operator represents a momentum-dependent term, which yields relatively minor contributions.

Term$[11k_1]$:

$$
[11k_1] = \left\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \right\|^{J_{ab}} \boldsymbol{\sigma}_l \cdot \hat{\boldsymbol{r}} \hat{\boldsymbol{r}} \left(1 + \frac{2}{m_\pi r} + \frac{2}{m_\pi^2 r^2} \right) Y_0(r) \boldsymbol{\tau}_\times^- \left\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \right\rangle^{J_{a'b'}}
$$

$$\xlongequal[(A-6)]{(A-4)(A-5)} \left\langle \tau_a\tau_b \middle| \boldsymbol{\tau}_\times^- \middle| \tau_{a'}\tau_{b'} \right\rangle \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}$$
$$\cdot \left\langle [NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \right\|^{J_{ab}} \boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\left(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2 r^2}\right)Y_0(\sqrt{2}r)\left\| [N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle^{J_{a'b'}} \tag{2-100}$$

$$\xlongequal[(B-15)]{(B-43)} 4i\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$
$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}$$
$$\cdot \left\langle [NL\otimes nl]^{L_{ab}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \right\|^{J_{ab}} \left[\boldsymbol{\sigma}_l\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\right]^1\left(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2 r^2}\right)Y_0(\sqrt{2}r)\left\| [N'L'\otimes n'l']^{L_{a'b'}}[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle^{J_{a'b'}} \tag{2-101}$$

$$\xlongequal{(A-11)} 4i\left(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\right)$$
$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'N'L'} \left\langle\!\left\langle NL, nl \middle| n_a l_a, n_b l_b \right\rangle\!\right\rangle^{L_{ab}} \left\langle\!\left\langle N'L', n'l' \middle| n_{a'} l_{a'}, n_{b'} l_{b'} \right\rangle\!\right\rangle^{L_{a'b'}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}$$
$$\cdot (-)^h\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} \left\langle [\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \middle\| \boldsymbol{\sigma}_l \middle\| [\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle$$
$$\cdot \left\langle [NL\otimes nl]^{L_{ab}} \middle\| \left(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2 r^2}\right)Y_0(\sqrt{2}r)[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h \middle\| [N'L'\otimes n'l']^{L_{a'b'}} \right\rangle \tag{2-102}$$

$$\frac{(B-24)}{(B-2)} 4i\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{L}_{ab}\hat{S}_{ab}\hat{L}_{a'b'}\hat{S}_{a'b'} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'N'L'} \Big\langle\!\Big\langle NL, nl \Big| n_a l_a, n_b l_b \Big\rangle\!\Big\rangle^{L_{ab}} \Big\langle\!\Big\langle N'L', n'l' \Big| n_{a'} l_{a'}, n_{b'} l_{b'} \Big\rangle\!\Big\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N'+L'+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \frac{1}{\sqrt{3}} \sum_{h=0}^{2} (-)^{h+1}\hat{h}$$

$$\cdot (-)^h\sqrt{3}\hat{J}_{ab}\hat{J}_{a'b'} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} (-)^{S_{ab}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix}$$

$$\cdot \delta_{NN'}\delta_{LL'}(-)^{h+l'+L'+L_{ab}} C^{h0}_{1010} C^{l0}_{l'0h0} \hat{l}' \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix}$$

$$\cdot \Big\langle \underbrace{nl}_{r} \Big| \Big(1 + \frac{\sqrt{2}}{m_\pi r} + \frac{1}{m_\pi^2 r^2}\Big) Y_0(\sqrt{2}r) \Big| n'l' \Big\rangle \tag{2-103}$$

$$= \Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^1 4\sqrt{6}\, i\, \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{ab}} \hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \Big\langle\!\Big\langle NL, nl \Big| n_a l_a, n_b l_b \Big\rangle\!\Big\rangle^{L_{ab}} \Big\langle\!\Big\langle NL, n'l' \Big| n_{a'} l_{a'}, n_{b'} l_{b'} \Big\rangle\!\Big\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} (-)^{l'+L}\hat{l}' \Big\langle \underbrace{nl}_{r} \Big| \Big(1 + \frac{\sqrt{2}}{m_\pi r} + \frac{1}{m_\pi^2 r^2}\Big) Y_0(\sqrt{2}r) \Big| n'l' \Big\rangle$$

$$\cdot \sum_{h=0}^{2} (-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010} C^{l0}_{l'0h0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-104}$$

In the code：

$$[11k_1] = \frac{\sqrt{6}\, g_A m_\pi^2}{2m_N F_\pi^2} \Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big) \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{ab}} \hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_a l_a, n_b l_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'} l_{a'}, n_{b'} l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|\Big(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2 r^2}\Big)Y_0(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h\hat{h}\begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0}C_{l'0h0}^{l0}\begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-105}$$

Term$[12L_1]$: (The derivation follows the same method as Term $[11K_1]$)

$$[12L_1] = \Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b\Big\|^{J_{ab}} \boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\frac{1}{r^2}Y_1(r)\boldsymbol{\tau}_\times^-\Big\|n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'}\Big\rangle^{J_{a'b'}}$$

$$= \Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^1 2\sqrt{6}\,i\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}+S_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_a l_a, n_b l_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'} l_{a'}, n_{b'} l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|\frac{1}{r^2}Y_1(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h\hat{h}\begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0}C_{l'0h0}^{l0}\begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-106}$$

In the code:

$$[12L_1] = \frac{\sqrt{6}\,g_A}{4m_N F_\pi^2}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}+S_{ab}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2\begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix}\begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix}\begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL,nl|n_a l_a, n_b l_b\rangle\rangle^{L_{ab}} \langle\langle NL,n'l'|n_{a'} l_{a'}, n_{b'} l_{b'}\rangle\rangle^{L_{a'b'}}$$

$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r}\Big|\frac{1}{r^2}Y_1(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h\hat{h}\begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0}C_{l'0h0}^{l0}\begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-107}$$

Term$[13M_1]$: (The derivation follows the same method as Term $[01A_1]$)

$$[13M_1]=\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_l\frac{1}{r^2}Y_1(r)\boldsymbol{\tau}_\times^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}}$$
$$=\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^{1+J_{ab}}\,2\sqrt{6}\,i\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix}S_{a'b'}&S_{ab}&1\\\frac{1}{2}&\frac{1}{2}&\frac{1}{2}\end{Bmatrix}\begin{Bmatrix}J_{a'b'}&J_{ab}&1\\S_{ab}&S_{a'b'}&L_{ab}\end{Bmatrix}\begin{Bmatrix}l_a&l_b&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\j_a&j_b&J_{ab}\end{Bmatrix}\begin{Bmatrix}l_{a'}&l_{b'}&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\\j_{a'}&j_{b'}&J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'}\Big\langle\Big\langle NL,nl\Big|n_al_a,n_bl_b\Big\rangle\Big\rangle^{L_{ab}}\Big\langle\Big\langle NL,n'l\Big|n_{a'}l_{a'},n_{b'}l_{b'}\Big\rangle\Big\rangle^{L_{ab}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle\underbrace{nl}_{r}\Big|\frac{1}{r^2}Y_1(\sqrt{2}r)\Big|n'l\Big\rangle\quad(2\text{-}108)$$

In the code:

$$[13M_1]=\frac{\sqrt{6}\,g_A}{4m_NF_\pi^2}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^{1+J_{ab}}\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$
$$\cdot\sum_{L_{ab}S_{ab}S_{a'b'}}(-)^{L_{ab}+S_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{S}_{a'b'}^2\begin{Bmatrix}S_{a'b'}&S_{ab}&1\\\frac{1}{2}&\frac{1}{2}&\frac{1}{2}\end{Bmatrix}\begin{Bmatrix}J_{a'b'}&J_{ab}&1\\S_{ab}&S_{a'b'}&L_{ab}\end{Bmatrix}\begin{Bmatrix}l_a&l_b&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\j_a&j_b&J_{ab}\end{Bmatrix}\begin{Bmatrix}l_{a'}&l_{b'}&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\\j_{a'}&j_{b'}&J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'}\Big\langle\Big\langle NL,nl\Big|n_al_a,n_bl_b\Big\rangle\Big\rangle^{L_{ab}}\Big\langle\Big\langle NL,n'l\Big|n_{a'}l_{a'},n_{b'}l_{b'}\Big\rangle\Big\rangle^{L_{ab}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}\Big\langle\underbrace{nl}_{r}\Big|\frac{1}{r^2}Y_1(\sqrt{2}r)\Big|n'l\Big\rangle\quad(2\text{-}109)$$

Term$[11k_2]$: (This term corresponds to the exchange term of $[11k_1](k\longleftrightarrow l)$)

$$[11k_2]=-\Big\langle n_al_aj_a\tau_a,n_bl_bj_b\tau_b\Big\|^{J_{ab}}\boldsymbol{\sigma}_k\cdot\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\Big(1+\frac{2}{m_\pi r}+\frac{2}{m_\pi^2r^2}\Big)Y_0(r)\boldsymbol{\tau}_\times^-\Big\|n_{a'}l_{a'}j_{a'}\tau_{a'},n_{b'}l_{b'}j_{b'}\tau_{b'}\Big\rangle^{J_{a'b'}}$$
$$=\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b-\delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)4\sqrt{6}\,i\,\hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$
$$\cdot\sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}}(-)^{L_{ab}+S_{a'b'}}\hat{L}_{ab}^2\hat{S}_{ab}^2\hat{L}_{a'b'}^2\hat{S}_{a'b'}^2\begin{Bmatrix}S_{a'b'}&S_{ab}&1\\\frac{1}{2}&\frac{1}{2}&\frac{1}{2}\end{Bmatrix}\begin{Bmatrix}l_a&l_b&L_{ab}\\\frac{1}{2}&\frac{1}{2}&S_{ab}\\j_a&j_b&J_{ab}\end{Bmatrix}\begin{Bmatrix}l_{a'}&l_{b'}&L_{a'b'}\\\frac{1}{2}&\frac{1}{2}&S_{a'b'}\\j_{a'}&j_{b'}&J_{a'b'}\end{Bmatrix}$$
$$\cdot\sum_{nlNL,n'l'}\Big\langle\Big\langle NL,nl\Big|n_al_a,n_bl_b\Big\rangle\Big\rangle^{L_{ab}}\Big\langle\Big\langle NL,n'l'\Big|n_{a'}l_{a'},n_{b'}l_{b'}\Big\rangle\Big\rangle^{L_{a'b'}}$$
$$\cdot\,\delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle\underbrace{nl}_{r}\Big|\Big(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2r^2}\Big)Y_0(\sqrt{2}r)\Big|n'l'\Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010} C^{l0}_{l'0h0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-110}$$

In the code:

$$[11k_2] = \frac{\sqrt{6}\, g_A m_\pi^2}{2m_N F_\pi^2}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^1\, \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r} \Big| \Big(1+\frac{\sqrt{2}}{m_\pi r}+\frac{1}{m_\pi^2 r^2}\Big) Y_0(\sqrt{2}r) \Big| n'l' \Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010} C^{l0}_{l'0h0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-111}$$

Term$[12L_2]$: (This term corresponds to the exchange term of $[12L_1](k \longleftrightarrow l)$)

$$[12L_2] = -\Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \Big\|^{J_{ab}} \boldsymbol{\sigma}_k \cdot \hat{\boldsymbol{r}}\hat{\boldsymbol{r}} \frac{1}{r^2} Y_1(r) \boldsymbol{\tau}^-_\times \Big\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \Big\rangle^{J_{a'b'}}$$

$$= \Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big) 2\sqrt{6}\, i\, \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}^2_{ab}\hat{S}^2_{ab}\hat{L}^2_{a'b'}\hat{S}^2_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$

$$\cdot \sum_{nlNL,n'l'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}$$

$$\cdot\, \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b}\delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}}(-)^{l'+L}\hat{l}'\Big\langle \underbrace{nl}_{r} \Big| \frac{1}{r^2} Y_1(\sqrt{2}r) \Big| n'l' \Big\rangle$$

$$\cdot \sum_{h=0}^{2}(-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C^{h0}_{1010} C^{l0}_{l'0h0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-112}$$

In the code:

$$[12L_2] = \frac{\sqrt{6}\, g_A}{4m_N F_\pi^2}\Big(\delta_{\tau_a,-\frac{1}{2}}\delta_{\tau_{a'},\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b - \delta_{\tau_b,-\frac{1}{2}}\delta_{\tau_{b'},\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big)(-)^1\, \hat{j}_a\hat{j}_b\hat{j}_{a'}\hat{j}_{b'}\hat{J}_{ab}\hat{J}_{a'b'}$$

$$\cdot \sum_{L_{ab}S_{ab}L_{a'b'}S_{a'b'}} (-)^{L_{ab}+S_{a'b'}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{L}_{a'b'}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{a'b'} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'l'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l' | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{a'b'}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l',2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} (-)^{l'+L} \hat{l}' \Big\langle \underbrace{nl}_{r} \Big| \frac{1}{r^2} Y_1(\sqrt{2}r) \Big| n'l' \Big\rangle$$
$$\cdot \sum_{h=0}^{2} (-)^h \hat{h} \begin{Bmatrix} h & 1 & 1 \\ L_{ab} & S_{ab} & J_{ab} \\ L_{a'b'} & S_{a'b'} & J_{a'b'} \end{Bmatrix} C_{1010}^{h0} C_{l'0h0}^{l0} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L \end{Bmatrix} \tag{2-113}$$

Term$[13M_2]$: (This term corresponds to the exchange term of $[13M_1](k \longleftrightarrow l)$)

$$[13M_2] = -\Big\langle n_a l_a j_a \tau_a, n_b l_b j_b \tau_b \Big\| \Big\|^{J_{ab}} \boldsymbol{\sigma}_k \frac{1}{r^2} Y_1(r) \boldsymbol{\tau}_\times^- \Big\| n_{a'} l_{a'} j_{a'} \tau_{a'}, n_{b'} l_{b'} j_{b'} \tau_{b'} \Big\rangle^{J_{a'b'}}$$
$$= \Big(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a\Big) (-)^{J_{ab}}\, 2\sqrt{6}\, i\, \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$
$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| \frac{1}{r^2} Y_1(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-114}$$

In the code:

$$[13M_2] = \frac{\sqrt{6}\, g_A}{4 m_N F_\pi^2} \Big(\delta_{\tau_a,-\frac{1}{2}} \delta_{\tau_{a'},\frac{1}{2}} \delta_{\tau_b,\tau_{b'}} \tau_b - \delta_{\tau_b,-\frac{1}{2}} \delta_{\tau_{b'},\frac{1}{2}} \delta_{\tau_a,\tau_{a'}} \tau_a\Big) (-)^{J_{ab}} \hat{j}_a \hat{j}_b \hat{j}_{a'} \hat{j}_{b'} \hat{J}_{ab} \hat{J}_{a'b'}$$
$$\cdot \sum_{L_{ab}S_{ab}S_{a'b'}} (-)^{L_{ab}} \hat{L}_{ab}^2 \hat{S}_{ab}^2 \hat{S}_{a'b'}^2 \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \begin{Bmatrix} J_{a'b'} & J_{ab} & 1 \\ S_{ab} & S_{a'b'} & L_{ab} \end{Bmatrix} \begin{Bmatrix} l_a & l_b & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ j_a & j_b & J_{ab} \end{Bmatrix} \begin{Bmatrix} l_{a'} & l_{b'} & L_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \\ j_{a'} & j_{b'} & J_{a'b'} \end{Bmatrix}$$
$$\cdot \sum_{nlNL,n'} \langle\langle NL, nl | n_a l_a, n_b l_b \rangle\rangle^{L_{ab}} \langle\langle NL, n'l | n_{a'} l_{a'}, n_{b'} l_{b'} \rangle\rangle^{L_{ab}}$$
$$\cdot \delta_{2N+L+2n+l,2n_a+l_a+2n_b+l_b} \delta_{2N+L+2n'+l,2n_{a'}+l_{a'}+2n_{b'}+l_{b'}} \Big\langle \underbrace{nl}_{r} \Big| \frac{1}{r^2} Y_1(\sqrt{2}r) \Big| n'l \Big\rangle \tag{2-115}$$

Chapter3 $2\nu\beta\beta$ Nuclear Matrix Element

For the $2\nu\beta\beta$ decay NMEs, because of isospin symmetry, Fermi transition is highly suppressed so that only the NME of the GT transition, $M_{GT}^{2\nu}$, is considered in calculations. The GT NME, in the QRPA approach is expressed as [2, 6],

$$M^{2\nu} \approx M_{GT}^{2\nu} = \sum_{if} \frac{\left\langle 0_f^+ \left\| O_{GT}^{(-)} \right\| 1_f^+ \right\rangle \left\langle 1_f^+ | 1_i^+ \right\rangle \left\langle 1_i^+ \left\| O_{GT}^{(-)} \right\| 0_i^+ \right\rangle}{\frac{1}{2}\left(\omega_i + \omega_f\right)}, \tag{3-1}$$

$$\left\langle 0_f^+ \left| O_{GT}^{(-)} \right| 1_f^+ \right\rangle = \left\langle 0_f^+ \left| \left[O_{GT}^{(-)}, \mathcal{Q}_f^\dagger(JM) \right] \right| 0_f^+ \right\rangle \tag{3-2}$$

$$\left\langle 1_i^+ \left| O_{GT}^{(-)} \right| 0_i^+ \right\rangle = \left\langle 0_i^+ \left| \left[\mathcal{Q}_f(JM), O_{GT}^{(-)} \right] \right| 0_i^+ \right\rangle \tag{3-3}$$

where, the energies ω_i and ω_f in the denominator correspond to the excitation energies relative to the ground states of initial and final nuclei.

After including the contribution of the GT two-body currents, the GT operator takes the form of a sum of one-body and two-body current components,

$$O_{GT} = O_{GT}^{1b} + O_{GT}^{2b}. \tag{3-4}$$

In Eq. (3-1), $\left\langle 1_f^+ | 1_i^+ \right\rangle$ is the overlap factor, given by the following expression,

$$\begin{aligned} \left\langle 1_f^+ | 1_i^+ \right\rangle = & \sum_{\pi_i \nu_i \pi_f \nu_f} \left\langle \pi_i | \pi_f \right\rangle \left\langle \nu_i | \nu_f \right\rangle \left(X_{\pi_i \nu_i}^i X_{\pi_f \nu_f}^f - Y_{\pi_i \nu_i}^i Y_{\pi_f \nu_f}^f \right) \\ & \cdot \left(u_{\pi_i} u_{\pi_f} + v_{\pi_i} v_{\pi_f} \right) \left(u_{\nu_i} u_{\nu_f} + v_{\nu_i} v_{\nu_f} \right) \left\langle HFB_f | HFB_i \right\rangle \end{aligned} \tag{3-5}$$

$$\left\langle HFB_f | HFB_i \right\rangle = \prod_{\pi_i \pi_f \nu_i \nu_f} \left(u_{\pi_i} u_{\pi_f} + v_{\pi_i} v_{\pi_f} \right) \left(u_{\nu_i} u_{\nu_f} + v_{\nu_i} v_{\nu_f} \right) \tag{3-6}$$

在QRPA理论中，初态核和末态核基态激发的中间态$|1_i^+\rangle$和$|1_f^+\rangle$不是正交的，因此需要计算其重叠因子。下面推导重叠因子 $\left\langle 1_f^+ | 1_i^+ \right\rangle$：

$$\left\langle 1_f^+ | 1_i^+ \right\rangle = \left\langle 0_f^+ \left| \mathcal{Q}_f(JM) \mathcal{Q}_i^\dagger(JM) \right| 0_i^+ \right\rangle = \left\langle QRPA_f \left| \mathcal{Q}_f(JM) \mathcal{Q}_i^\dagger(JM) \right| QRPA_i \right\rangle \tag{3-7}$$

将初态核的QRPA激发算符在末态核的QRPA激发算符上展开：

$$\mathcal{Q}_i^\dagger(JM) = \sum_f \left(a_{if} \mathcal{Q}_f^\dagger(JM) + b_{if} \tilde{\mathcal{Q}}_f(JM) \right) \tag{3-8}$$

将公式(3-8)代入公式(3-7)中：

$$\left\langle 1_f^+ | 1_i^+ \right\rangle = \left\langle QRPA_f \left| \mathcal{Q}_f(JM) \sum_f \left(a_{if} \mathcal{Q}_f^\dagger(JM) + b_{if} \tilde{\mathcal{Q}}_f(JM) \right) \right| QRPA_i \right\rangle$$

$$
\begin{aligned}
&= \sum_f \left[\left\langle QRPA_f \left| a_{if} \mathcal{Q}_f(JM) \mathcal{Q}_f^\dagger(JM) \right| QRPA_i \right\rangle + \left\langle QRPA_f \left| b_{if} \mathcal{Q}_f(JM) \tilde{\mathcal{Q}}_f(JM) \right| QRPA_i \right\rangle \right] \\
&\approx \sum_f a_{if} \left\langle QRPA_f | QRPA_i \right\rangle \overset{QBA}{\approx} \sum_f a_{if} \left\langle HFB_f | HFB_i \right\rangle \qquad (3\text{-}9)
\end{aligned}
$$

其中，忽略掉b_{if}项是因为初态核的QRPA基态和由末态核的QRPA基态激发的双声子激发态之间的重叠因子很小。

初态核和末态核的粒子算符通过单粒子态重叠矩阵联系：

$$
c_\alpha^{(i)\dagger} = \sum_\beta \left\langle \alpha | \beta \right\rangle c_\beta^{(f)\dagger} \qquad (3\text{-}10)
$$

其中，单粒子态$|\alpha\rangle$中的好量子数包含(n, l, j, m, τ)。

QRPA的激发算符表达式为：

$$
\mathcal{Q}_i^\dagger(JM) = \sum_{\pi_i \nu_i} X_{\pi_i \nu_i}^{iJ} A_{\pi_i \nu_i}^\dagger(JM) - Y_{\pi_i \nu_i}^{iJ} \tilde{A}_{\pi_i \nu_i}(JM) \qquad (3\text{-}11)
$$

因此：

$$
\begin{aligned}
& A_{\pi_i \nu_i}^\dagger(JM) = \sum_{m_{\pi_i} m_{\nu_i}} C_{j_{\pi_i} m_{\pi_i} j_{\nu_i} m_{\nu_i}}^{JM} a_{\pi_i}^\dagger a_{\nu_i}^\dagger \\
&= \sum_{m_{\pi_i} m_{\nu_i}} C_{j_{\pi_i} m_{\pi_i} j_{\nu_i} m_{\nu_i}}^{JM} (u_{\pi_i} c_{\pi_i}^\dagger - v_{\pi_i} \tilde{c}_{\pi_i})(u_{\nu_i} c_{\nu_i}^\dagger - v_{\nu_i} \tilde{c}_{\nu_i}) \\
&= \sum_{m_{\pi_i} m_{\nu_i}} C_{j_{\pi_i} m_{\pi_i} j_{\nu_i} m_{\nu_i}}^{JM} \sum_{\pi_f} \left\langle \pi_i | \pi_f \right\rangle (u_{\pi_i} c_{\pi_f}^\dagger - v_{\pi_i} \tilde{c}_{\pi_f}) \sum_{\nu_f} \left\langle \nu_i | \nu_f \right\rangle (u_{\nu_i} c_{\nu_f}^\dagger - v_{\nu_i} \tilde{c}_{\nu_f}) \\
&= \sum_{m_{\pi_i} m_{\nu_i}} C_{j_{\pi_i} m_{\pi_i} j_{\nu_i} m_{\nu_i}}^{JM} \sum_{\pi_f \nu_f} \left\langle \pi_i | \pi_f \right\rangle \left\langle \nu_i | \nu_f \right\rangle \left[u_{\pi_i} (u_{\pi_f} a_{\pi_f}^\dagger + v_{\pi_f} \tilde{a}_{\pi_f}) - v_{\pi_i} (u_{\pi_f} \tilde{a}_{\pi_f} - v_{\pi_f} a_{\pi_f}^\dagger) \right] \\
&\quad \cdot \left[u_{\nu_i} (u_{\nu_f} a_{\nu_f}^\dagger + v_{\nu_f} \tilde{a}_{\nu_f}) - v_{\nu_i} (u_{\nu_f} \tilde{a}_{\nu_f} - v_{\nu_f} a_{\nu_f}^\dagger) \right] \\
&= \sum_{m_{\pi_i} m_{\nu_i}} C_{j_{\pi_i} m_{\pi_i} j_{\nu_i} m_{\nu_i}}^{JM} \sum_{\pi_f \nu_f} \left\langle \pi_i | \pi_f \right\rangle \left\langle \nu_i | \nu_f \right\rangle \left[(u_{\pi_i} u_{\pi_f} + v_{\pi_i} v_{\pi_f}) a_{\pi_f}^\dagger + (u_{\pi_i} v_{\pi_f} - v_{\pi_i} u_{\pi_f}) \tilde{a}_{\pi_f} \right] \\
&\quad \cdot \left[(u_{\nu_i} u_{\nu_f} + v_{\nu_i} v_{\nu_f}) a_{\nu_f}^\dagger + (u_{\nu_i} v_{\nu_f} - v_{\nu_i} u_{\nu_f}) \tilde{a}_{\nu_f} \right] \\
&= \sum_{m_{\pi_f} m_{\nu_f}} C_{j_{\pi_f} m_{\pi_f} j_{\nu_f} m_{\nu_f}}^{JM} \sum_{\pi_f \nu_f} \left\langle \pi_i | \pi_f \right\rangle \left\langle \nu_i | \nu_f \right\rangle \\
&\quad \cdot \left[(u_{\pi_i} u_{\pi_f} + v_{\pi_i} v_{\pi_f})(u_{\nu_i} u_{\nu_f} + v_{\nu_i} v_{\nu_f}) a_{\pi_f}^\dagger a_{\nu_f}^\dagger + (u_{\pi_i} v_{\pi_f} - v_{\pi_i} u_{\pi_f})(u_{\nu_i} v_{\nu_f} - v_{\nu_i} u_{\nu_f}) \tilde{a}_{\pi_f} \tilde{a}_{\nu_f} \right] \\
&= \sum_{\pi_f \nu_f} \left\langle \pi_i | \pi_f \right\rangle \left\langle \nu_i | \nu_f \right\rangle \Bigg[(u_{\pi_i} u_{\pi_f} + v_{\pi_i} v_{\pi_f})(u_{\nu_i} u_{\nu_f} + v_{\nu_i} v_{\nu_f}) \sum_{m_{\pi_f} m_{\nu_f}} C_{j_{\pi_f} m_{\pi_f} j_{\nu_f} m_{\nu_f}}^{JM} a_{\pi_f}^\dagger a_{\nu_f}^\dagger \\
&\quad + (u_{\pi_i} v_{\pi_f} - v_{\pi_i} u_{\pi_f})(u_{\nu_i} v_{\nu_f} - v_{\nu_i} u_{\nu_f})
\end{aligned}
$$

$$
\begin{aligned}
&\cdot \sum_{m_{\pi_f} m_{\nu_f}} (-)^{j_{\pi_f}+j_{\nu_f}-J} C^{J-M}_{j_{\pi_f}-m_{\pi_f} j_{\nu_f}-m_{\nu_f}} (-)^{j_{\pi_f}+m_{\pi_f}} a_{-\pi_f} (-)^{j_{\nu_f}+m_{\nu_f}} a_{-\nu_f} \Bigg] \\
=&\sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Bigg[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f}) \sum_{m_{\pi_f} m_{\nu_f}} C^{JM}_{j_{\pi_f}m_{\pi_f}j_{\nu_f}m_{\nu_f}} a^\dagger_{\pi_f} a^\dagger_{\nu_f} \\
&+(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f}) \sum_{m_{\pi_f} m_{\nu_f}} (-)^{J+M} C^{J-M}_{j_{\pi_f}-m_{\pi_f} j_{\nu_f}-m_{\nu_f}} a_{-\pi_f} a_{-\nu_f} \Bigg] \\
=&\sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f}) A^\dagger_{\pi_f\nu_f}(JM) \\
&-(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f}) \tilde{A}_{\pi_f\nu_f}(JM) \Big]
\end{aligned}
\tag{3-12}
$$

同理，

$$
\begin{aligned}
&\tilde{A}_{\pi_i\nu_i}(JM) = (-)^{J+M} \sum_{m_{\pi_i} m_{\nu_i}} C^{J-M}_{j_{\pi_i}m_{\pi_i}j_{\nu_i}m_{\nu_i}} a_{\nu_i} a_{\pi_i} \\
=&(-)^{J+M} \sum_{m_{\pi_i} m_{\nu_i}} C^{J-M}_{j_{\pi_i}m_{\pi_i}j_{\nu_i}m_{\nu_i}} (u_{\nu_i}c_{\nu_i} - v_{\nu_i}\tilde{c}^\dagger_{\nu_i})(u_{\pi_i}c_{\pi_i} - v_{\pi_i}\tilde{c}^\dagger_{\pi_i}) \\
=&(-)^{J+M} \sum_{m_{\pi_i} m_{\nu_i}} C^{J-M}_{j_{\pi_i}m_{\pi_i}j_{\nu_i}m_{\nu_i}} \sum_{\nu_f} \langle\nu_i|\nu_f\rangle (u_{\nu_i}c_{\nu_f} - v_{\nu_i}\tilde{c}^\dagger_{\nu_f}) \sum_{\pi_f} \langle\pi_i|\pi_f\rangle (u_{\pi_i}c_{\pi_f} - v_{\pi_i}\tilde{c}^\dagger_{\pi_f}) \\
=&(-)^{J+M} \sum_{m_{\pi_i} m_{\nu_i}} C^{J-M}_{j_{\pi_i}m_{\pi_i}j_{\nu_i}m_{\nu_i}} \sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Big[u_{\nu_i}(u_{\nu_f}a_{\nu_f} + v_{\nu_f}\tilde{a}^\dagger_{\nu_f}) - v_{\nu_i}(u_{\nu_f}\tilde{a}^\dagger_{\nu_f} - v_{\nu_f}a_{\nu_f}) \Big] \\
&\cdot \Big[u_{\pi_i}(u_{\pi_f}a_{\pi_f} + v_{\pi_f}\tilde{a}^\dagger_{\pi_f}) - v_{\pi_i}(u_{\pi_f}\tilde{a}^\dagger_{\pi_f} - v_{\pi_f}a_{\pi_f}) \Big] \\
=&(-)^{J+M} \sum_{m_{\pi_i} m_{\nu_i}} C^{J-M}_{j_{\pi_i}m_{\pi_i}j_{\nu_i}m_{\nu_i}} \sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})a_{\nu_f} + (u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})\tilde{a}^\dagger_{\nu_f} \Big] \\
&\cdot \Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})a_{\pi_f} + (u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})\tilde{a}^\dagger_{\pi_f} \Big] \\
=&(-)^{J+M} \sum_{m_{\pi_f} m_{\nu_f}} C^{J-M}_{j_{\pi_f}m_{\pi_f}j_{\nu_f}m_{\nu_f}} \sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \\
&\cdot \Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})a_{\nu_f}a_{\pi_f} + (u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})\tilde{a}^\dagger_{\nu_f}\tilde{a}^\dagger_{\pi_f} \Big] \\
=&\sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Bigg[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(-)^{J+M} \sum_{m_{\pi_f} m_{\nu_f}} C^{J-M}_{j_{\pi_f}m_{\pi_f}j_{\nu_f}m_{\nu_f}} a_{\nu_f}a_{\pi_f} \\
&+(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f}) \\
&\cdot(-)^{J+M} \sum_{m_{\pi_f} m_{\nu_f}} (-)^{j_{\pi_f}+j_{\nu_f}-J} C^{JM}_{j_{\pi_f}-m_{\pi_f}j_{\nu_f}-m_{\nu_f}} (-)^{j_{\nu_f}+m_{\nu_f}} a^\dagger_{-\nu_f} (-)^{j_{\pi_f}+m_{\pi_f}} a^\dagger_{-\pi_f} \Bigg] \\
=&\sum_{\pi_f\nu_f} \langle\pi_i|\pi_f\rangle \langle\nu_i|\nu_f\rangle \Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f}) \tilde{A}_{\pi_f\nu_f}(JM)
\end{aligned}
$$

$$-(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})A^{\dagger}_{\pi_f\nu_f}(JM)\Big] \tag{3-13}$$

因为：

$$\mathcal{Q}^{\dagger}_f(JM)=\sum_{\pi_f\nu_f}X^{fJ}_{\pi_f\nu_f}A^{\dagger}_{\pi_f\nu_f}(JM)-Y^{fJ}_{\pi_f\nu_f}\tilde{A}_{\pi_f\nu_f}(JM) \tag{3-14}$$

$$\begin{aligned}\tilde{\mathcal{Q}}_f(JM)&=(-)^{J+M}\mathcal{Q}_f(J-M)\\&=(-)^{J+M}\sum_{\pi_f\nu_f}X^{fJ}_{\pi_f\nu_f}A_{\pi_f\nu_f}(J-M)-Y^{fJ}_{\pi_f\nu_f}\tilde{A}^{\dagger}_{\pi_f\nu_f}(J-M)\\&=\sum_{\pi_f\nu_f}X^{fJ}_{\pi_f\nu_f}\tilde{A}_{\pi_f\nu_f}(JM)-Y^{fJ}_{\pi_f\nu_f}A^{\dagger}_{\pi_f\nu_f}(JM)\end{aligned} \tag{3-15}$$

将公式(3-14)和公式(3-15)写成矩阵的形式：

$$\begin{pmatrix}\mathcal{Q}^{\dagger}_f\\ \tilde{\mathcal{Q}}_f\end{pmatrix}=\begin{pmatrix}X & -Y\\ -Y & X\end{pmatrix}\begin{pmatrix}A^{\dagger}_{\pi_f\nu_f}\\ \tilde{A}_{\pi_f\nu_f}\end{pmatrix} \tag{3-16}$$

对其做一个幺正变换：

$$\begin{aligned}&\begin{pmatrix}X & Y\\ Y & X\end{pmatrix}\begin{pmatrix}\mathcal{Q}^{\dagger}_f\\ \tilde{\mathcal{Q}}_f\end{pmatrix}=\begin{pmatrix}X & Y\\ Y & X\end{pmatrix}\begin{pmatrix}X & -Y\\ -Y & X\end{pmatrix}\begin{pmatrix}A^{\dagger}_{\pi_f\nu_f}\\ \tilde{A}_{\pi_f\nu_f}\end{pmatrix}\\ \Longrightarrow&\begin{pmatrix}X^2-Y^2 & 0\\ 0 & X^2-Y^2\end{pmatrix}\begin{pmatrix}A^{\dagger}_{\pi_f\nu_f}\\ \tilde{A}_{\pi_f\nu_f}\end{pmatrix}=\begin{pmatrix}X & Y\\ Y & X\end{pmatrix}\begin{pmatrix}\mathcal{Q}^{\dagger}_f\\ \tilde{\mathcal{Q}}_f\end{pmatrix}\end{aligned} \tag{3-17}$$

由文献 [4] 中(19.9)式知QRPA方程满足的正交归一性：

$$\begin{pmatrix}A^{\dagger}_{\pi_f\nu_f}\\ \tilde{A}_{\pi_f\nu_f}\end{pmatrix}=\begin{pmatrix}X & Y\\ Y & X\end{pmatrix}\begin{pmatrix}\mathcal{Q}^{\dagger}_f\\ \tilde{\mathcal{Q}}_f\end{pmatrix} \tag{3-18}$$

因此：

$$A^{\dagger}_{\pi_f\nu_f}(JM)=\sum_f X^{fJ}_{\pi_f\nu_f}\mathcal{Q}^{\dagger}_f(JM)+Y^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM) \tag{3-19}$$

$$\tilde{A}_{\pi_f\nu_f}(JM)=\sum_f X^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM)+Y^{fJ}_{\pi_f\nu_f}\mathcal{Q}^{\dagger}_f(JM) \tag{3-20}$$

联立公式(3-12,3-13,3-19,3-20)得：

$$\begin{aligned}A^{\dagger}_{\pi_i\nu_i}(JM)&=\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})A^{\dagger}_{\pi_f\nu_f}(JM)\\&-(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})\tilde{A}_{\pi_f\nu_f}(JM)\Big]\\&=\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})\Big(\sum_f X^{fJ}_{\pi_f\nu_f}\mathcal{Q}^{\dagger}_f(JM)+Y^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM)\Big)\end{aligned}$$

$$
-(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})\left(\sum_f X^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM)+Y^{fJ}_{\pi_f\nu_f}\mathcal{Q}^\dagger_f(JM)\right)\Bigg]
\tag{3-21}
$$

$$
\begin{aligned}
\tilde{A}_{\pi_i\nu_i}(JM)=&\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})\tilde{A}_{\pi_f\nu_f}(JM)\\
&-(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})A^\dagger_{\pi_f\nu_f}(JM)\Big]\\
=&\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\Bigg[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})\left(\sum_f X^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM)+Y^{fJ}_{\pi_f\nu_f}\mathcal{Q}^\dagger_f(JM)\right)\\
&-(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})\left(\sum_f X^{fJ}_{\pi_f\nu_f}\mathcal{Q}^\dagger_f(JM)+Y^{fJ}_{\pi_f\nu_f}\tilde{\mathcal{Q}}_f(JM)\right)\Bigg]
\end{aligned}
\tag{3-22}
$$

所以：

$$
\begin{aligned}
\mathcal{Q}^\dagger_i(JM)\approx&\sum_{\pi_i\nu_i}X^{iJ}_{\pi_i\nu_i}A^\dagger_{\pi_i\nu_i}(JM)-Y^{iJ}_{\pi_i\nu_i}\tilde{A}_{\pi_i\nu_i}(JM)\\
=&\sum_{\pi_i\nu_i}X^{iJ}_{\pi_i\nu_i}\Bigg(\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\sum_f\Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})X^{fJ}_{\pi_f\nu_f}\\
&-(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})Y^{fJ}_{\pi_f\nu_f}\Big]\mathcal{Q}^\dagger_f(JM)\Bigg)\\
&-Y^{iJ}_{\pi_i\nu_i}\Bigg(\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\sum_f\Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})Y^{fJ}_{\pi_f\nu_f}\\
&-(u_{\nu_i}v_{\nu_f}-v_{\nu_i}u_{\nu_f})(u_{\pi_i}v_{\pi_f}-v_{\pi_i}u_{\pi_f})X^{fJ}_{\pi_f\nu_f}\Big]\mathcal{Q}^\dagger_f(JM)\Bigg)\\
\approx&\sum_{\pi_i\nu_i}X^{iJ}_{\pi_i\nu_i}\left(\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\sum_f\Big[(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})X^{fJ}_{\pi_f\nu_f}\Big]\mathcal{Q}^\dagger_f(JM)\right)\\
&-Y^{iJ}_{\pi_i\nu_i}\left(\sum_{\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle\sum_f\Big[(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})Y^{fJ}_{\pi_f\nu_f}\Big]\mathcal{Q}^\dagger_f(JM)\right)\\
=&\sum_f\left(\sum_{\pi_i\nu_i\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})\left(X^{iJ}_{\pi_i\nu_i}X^{fJ}_{\pi_f\nu_f}-Y^{iJ}_{\pi_i\nu_i}Y^{fJ}_{\pi_f\nu_f}\right)\right)\mathcal{Q}^\dagger_f(JM)
\end{aligned}
\tag{3-23}
$$

所以：

$$a_{if}=\sum_{\pi_i\nu_i\pi_f\nu_f}\langle\pi_i|\pi_f\rangle\langle\nu_i|\nu_f\rangle(u_{\pi_i}u_{\pi_f}+v_{\pi_i}v_{\pi_f})(u_{\nu_i}u_{\nu_f}+v_{\nu_i}v_{\nu_f})\left(X^{iJ}_{\pi_i\nu_i}X^{fJ}_{\pi_f\nu_f}-Y^{iJ}_{\pi_i\nu_i}Y^{fJ}_{\pi_f\nu_f}\right) \tag{3-24}$$

令$c_i^{k\dagger}=c_f^{k\dagger}$,有：

$$\begin{aligned}
\langle HFB_f|HFB_i\rangle &= \left\langle 0\left|\prod_k\left(u_f^k+v_f^k\tilde{c}_f^k c_f^k\right)\prod_l\left(u_i^l+v_i^l c_i^{l\dagger}\tilde{c}_i^{l\dagger}\right)\right|0\right\rangle\\
&=\left\langle 0\left|\prod_k\left(u_f^k+v_f^k\tilde{c}_f^k c_f^k\right)\prod_l\left(u_i^l+v_i^l c_f^{l\dagger}\tilde{c}_f^{l\dagger}\right)\right|0\right\rangle\\
&=\prod_k u_i^k u_f^k\\
&\quad+\sum_{k_1}v_i^{k_1}v_f^{k_1}\left\langle 0\left|\tilde{c}_f^{k_1}c_f^{k_1}c_f^{k_1\dagger}\tilde{c}_f^{k_1\dagger}\right|0\right\rangle\prod_{k\neq k_1}u_i^k u_f^k\\
&\quad+\sum_{k_1\neq k_2}v_i^{k_1}v_f^{k_1}v_i^{k_2}v_f^{k_2}\left\langle 0\left|\tilde{c}_f^{k_1}c_f^{k_1}\tilde{c}_f^{k_2}c_f^{k_2}c_f^{k_1\dagger}\tilde{c}_f^{k_1\dagger}c_f^{k_2\dagger}\tilde{c}_f^{k_2\dagger}\right|0\right\rangle\prod_{k\neq k_1,k_2}u_i^k u_f^k\\
&\quad+\cdots+\prod_k v_i^k v_f^k\\
&=\prod_k u_i^k u_f^k\\
&\quad+\sum_{k_1}v_i^{k_1}v_f^{k_1}\prod_{k\neq k_1}u_i^k u_f^k\\
&\quad+\sum_{k_1\neq k_2}v_i^{k_1}v_f^{k_1}v_i^{k_2}v_f^{k_2}\prod_{k\neq k_1,k_2}u_i^k u_f^k\\
&\quad+\cdots+\prod_k v_i^k v_f^k\\
&=\prod_k\left(u_i^k u_f^k+v_i^k v_f^k\right)
\end{aligned} \tag{3-25}$$

Appendix A

A.1 Basic formula

Define,$\tau_n = \frac{1}{2}, \tau_p = -\frac{1}{2}$, where n and p represent the neutron and proton, respectively.

$$\left\langle \tau_a \tau_b \middle| \boldsymbol{\tau}_l^{\pm} \middle| \tau_{a'} \tau_{b'} \right\rangle = \delta_{\tau_a \tau_{a'}} \delta_{\tau_b, \pm\frac{1}{2}} \delta_{\tau_{b'}, \mp\frac{1}{2}} \tag{A-1}$$

$$\left\langle \tau_a \tau_b \middle| \boldsymbol{\tau}_k^{\pm} \middle| \tau_{a'} \tau_{b'} \right\rangle = \delta_{\tau_b \tau_{b'}} \delta_{\tau_a, \pm\frac{1}{2}} \delta_{\tau_{a'}, \mp\frac{1}{2}} \tag{A-2}$$

$$\left\langle \frac{1}{2} \middle\| \boldsymbol{\sigma} \middle\| \frac{1}{2} \right\rangle \xlongequal[P31(2.39)]{[4]} \sqrt{6} \tag{A-3}$$

A.2 Formulas in textbooks and literature

$$\left| j_1 j_3 \left(j_{13} \right) j_2 j_4 \left(j_{24} \right) \right\rangle^{jm} \xlongequal[P16(1.84)]{[4]} \sum_{j_{12} j_{34}} \hat{j}_{12} \hat{j}_{34} \hat{j}_{13} \hat{j}_{24} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j_4 & j_{34} \\ j_{13} & j_{24} & j \end{Bmatrix} \left| j_1 j_2 \left(j_{12} \right) j_3 j_4 \left(j_{34} \right) \right\rangle^{jm} \tag{A-4}$$

MOSHINSKY Transformation（$\boldsymbol{r} = \frac{1}{\sqrt{2}}(\boldsymbol{r}_1 - \boldsymbol{r}_2)$）：

$$\left| n_1 l_1, n_2 l_2 \right\rangle^{\lambda\mu} \xlongequal[P192(1)]{[1]} \sum_{nlNL} \left| NL, nl \right\rangle^{\lambda\mu} \langle\langle NL, nl \middle| n_1 l_1, n_2 l_2 \rangle\rangle^{\lambda} \tag{A-5}$$

This transformation satisfies the following relations:

$$2n_1 + l_1 + 2n_2 + l_2 \xlongequal[P193(5)]{[1]} 2n + l + 2N + L \tag{A-6}$$

Reduced Matrix Elements for Different Operators:

$$\langle l \| 1 \| l \rangle = \hat{l} \tag{A-7}$$

A.3 Simple formulas derived from textbooks and literature

★ As given in Ref.[1] P479(28),

$$\langle n_1' j_1', n_2' j_2' |^{j'm'} [\hat{\mathbf{P}}_a(1) \otimes \hat{\mathbf{Q}}_b(2)]^{c\gamma} | n_1 j_1, n_2 j_2 \rangle^{jm}$$

$$= (-1)^{2c}\,\hat{c}\hat{j}C^{j'm'}_{jmc\gamma}\begin{Bmatrix} a & b & c \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\langle n_1'j_1'\|\hat{\mathbf{P}}_a(1)\|n_1j_1\rangle\langle n_2'j_2'\|\hat{\mathbf{Q}}_b(2)\|n_2j_2\rangle \tag{A-8}$$

$$\xRightarrow[P475(2)]{[5]} (-)^{2c}C^{j'm'}_{jmc\gamma}\frac{1}{\hat{j}'}\langle n_1'j_1', n_2'j_2'\|^{j'}[\hat{\mathbf{P}}_a(1)\otimes\hat{\mathbf{Q}}_b(2)]^c\|n_1j_1, n_2j_2\rangle^j$$

$$= (-1)^{2c}\,\hat{c}\hat{j}C^{j'm'}_{jmc\gamma}\begin{Bmatrix} a & b & c \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\langle n_1'j_1'\|\hat{\mathbf{P}}_a(1)\|n_1j_1\rangle\langle n_2'j_2'\|\hat{\mathbf{Q}}_b(2)\|n_2j_2\rangle \tag{A-9}$$

$$\Longrightarrow \langle n_1'j_1', n_2'j_2'\|^{j'}[\hat{\mathbf{P}}_a(1)\otimes\hat{\mathbf{Q}}_b(2)]^c\|n_1j_1, n_2j_2\rangle^j$$

$$= \hat{c}\hat{j}\hat{j}'\begin{Bmatrix} a & b & c \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\langle n_1'j_1'\|\hat{\mathbf{P}}_a(1)\|n_1j_1\rangle\langle n_2'j_2'\|\hat{\mathbf{Q}}_b(2)\|n_2j_2\rangle \tag{A-10}$$

★ As given in Ref.[8] P34(A-47),

$$\left\langle n_1'j_1', n_2'j_2'\right\|^{j'}[\hat{\mathbf{Q}}_b(2)\otimes\hat{\mathbf{P}}_a(1)]^c\left\| n_1j_1, n_2j_2\right\rangle^j$$

$$= (-)^{a+b-c}\hat{c}\hat{j}\hat{j}'\begin{Bmatrix} a & b & c \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\left\langle n_1'j_1'\left\|\hat{\mathbf{P}}_a(1)\right\| n_1j_1\right\rangle\left\langle n_2'j_2'\left\|\hat{\mathbf{Q}}_b(2)\right\| n_2j_2\right\rangle \tag{A-11}$$

★ As given in Ref.(A-10),

$$\langle j_1', j_2'\|^{j'}\hat{\mathbf{P}}_a(1)\|j_1, j_2\rangle^j$$

$$= \hat{a}\hat{j}\hat{j}'\begin{Bmatrix} a & 0 & a \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\langle j_1'\|\hat{\mathbf{P}}_a(1)\|j_1\rangle\langle j_2'\|1\|j_2\rangle \tag{A-12}$$

$$\xlongequal[P357(2)]{[5]} \delta_{j_2j_2'}(-)^{a+j_1'+j_2+j}\frac{\hat{j}\hat{j}'}{\hat{j}_2}\begin{Bmatrix} j' & j & a \\ j_1 & j_1' & j_2 \end{Bmatrix}\langle j_1'\|\hat{\mathbf{P}}_a(1)\|j_1\rangle\langle j_2\|1\|j_2\rangle \tag{A-13}$$

$$\xlongequal[P484(9)]{[5]} \delta_{j_2j_2'}(-)^{a+j_1'+j_2+j}\hat{j}\hat{j}'\begin{Bmatrix} j' & j & a \\ j_1 & j_1' & j_2 \end{Bmatrix}\langle j_1'\|\hat{\mathbf{P}}_a(1)\|j_1\rangle \tag{A-14}$$

★ As given in Ref.(A-10),

$$\langle j_1', j_2'\|^{j'}\hat{\mathbf{Q}}_b(2)\|j_1, j_2\rangle^j$$

$$= \hat{b}\hat{j}\hat{j}'\begin{Bmatrix} 0 & b & b \\ j_1' & j_2' & j' \\ j_1 & j_2 & j \end{Bmatrix}\langle j_1'\|1\|j_1\rangle\langle j_2'\|\hat{\mathbf{Q}}_b(2)\|j_2\rangle \tag{A-15}$$

$$\xlongequal[P357(2)]{[5]} \delta_{j_1 j_1'} (-)^{b+j_1+j_2+j'} \frac{\hat{j}\hat{j}'}{\hat{j}_1} \begin{Bmatrix} j & j' & b \\ j_2' & j_2 & j_1 \end{Bmatrix} \langle j_1 \| 1 \| j_1 \rangle \langle j_2' \| \hat{\mathbf{Q}}_b(2) \| j_2 \rangle \tag{A-16}$$

$$\xlongequal[P484(9)]{[5]} \delta_{j_1 j_1'} (-)^{b+j_1+j_2+j'} \hat{j}\hat{j}' \begin{Bmatrix} j & j' & b \\ j_2' & j_2 & j_1 \end{Bmatrix} \langle j_2' \| \hat{\mathbf{Q}}_b(2) \| j_2 \rangle \tag{A-17}$$

★ As given in Ref.[1]P165(18),

$$\delta(\boldsymbol{r}_1 - \boldsymbol{r}_2) = \frac{\delta(r_1 - r_2)}{r_1^2} \sum_{l=0}^{\infty} \left(\hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1) \cdot \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2) \right) \tag{A-18}$$

$$\xlongequal[P65(35)]{[5]} \frac{\delta(r_1 - r_2)}{r_1^2} \sum_{l=0}^{\infty} (-)^l \, \hat{l} \left[\hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1) \otimes \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2) \right]^0 \tag{A-19}$$

Appendix B

B.1 Derived formula

$$\bigstar \left\langle [\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \middle\| \boldsymbol{\sigma}_l \middle\| [\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle$$

$$\overset{(A-17)}{=\!=\!=} (-)^{S_{ab}}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \left\langle \frac{1}{2} \middle\| \boldsymbol{\sigma}_l \middle\| \frac{1}{2} \right\rangle \tag{B-1}$$

$$\overset{(A-3)}{=\!=\!=} (-)^{S_{ab}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \tag{B-2}$$

$$\bigstar \left\langle [\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}} \middle\| \boldsymbol{\sigma}_k \middle\| [\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}} \right\rangle \tag{B-3}$$

$$\overset{(A-14)}{=\!=\!=} (-)^{S_{a'b'}}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{ab} & S_{a'b'} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \left\langle \frac{1}{2} \middle\| \boldsymbol{\sigma}_l \middle\| \frac{1}{2} \right\rangle \tag{B-4}$$

$$\overset{(A-3)}{=\!=\!=} (-)^{S_{a'b'}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{ab} & S_{a'b'} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \tag{B-5}$$

$$= (-)^{S_{a'b'}}\sqrt{6}\hat{S}_{ab}\hat{S}_{a'b'} \begin{Bmatrix} S_{a'b'} & S_{ab} & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{Bmatrix} \tag{B-6}$$

$$\bigstar \left\langle [NL\otimes nl]^{L_{ab}} \middle\| Y_2(r) \middle\| [N'L'\otimes n'l']^{L_{a'b'}} \right\rangle$$

$$\overset{(A-17)}{=\!=\!=} \delta_{NN'}\delta_{LL'}(-)^{L'+l'+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{a'b'} & L_{ab} & 0 \\ l & l' & L' \end{Bmatrix} \left\langle \underbrace{nl}_{\boldsymbol{r}} \middle\| Y_2(r) \middle\| n'l' \right\rangle \tag{B-7}$$

$$= \delta_{NN'}\delta_{LL'}(-)^{L'+l'+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{a'b'} & L_{ab} & 0 \\ l & l' & L' \end{Bmatrix} \left\langle \underbrace{nl}_{r} \middle| Y_2(r) \middle| n'l' \right\rangle \left\langle \underbrace{l}_{\hat{\boldsymbol{r}}} \middle\| 1 \middle\| l' \right\rangle \tag{B-8}$$

$$\overset{(A-7)}{=\!=\!=} \delta_{ll'}\delta_{NN'}\delta_{LL'}(-)^{L'+l'+L_{ab}}\hat{l}\hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{a'b'} & L_{ab} & 0 \\ l & l' & L' \end{Bmatrix} \left\langle \underbrace{nl}_{r} \middle| Y_2(r) \middle| n'l' \right\rangle \tag{B-9}$$

$$\underset{P299(1)}{\overset{[5]}{=\!=\!=}} \delta_{ll'}\delta_{NN'}\delta_{LL'}(-)^{L'+l'+L_{ab}}\hat{l}\hat{L}_{ab}\hat{L}_{a'b'}(-)^{L_{ab}+l+L'}\delta_{L_{ab}L_{a'b'}}\frac{1}{\hat{L}_{ab}\hat{l}} \left\langle \underbrace{nl}_{r} \middle| Y_2(r) \middle| n'l' \right\rangle \tag{B-10}$$

$$= \delta_{ll'}\delta_{NN'}\delta_{LL'}\delta_{L_{ab}L_{a'b'}}\hat{L}_{ab} \left\langle \underbrace{nl}_{r} \middle| Y_2(r) \middle| n'l \right\rangle \tag{B-11}$$

$$\bigstar\ \boldsymbol{\sigma}_l \cdot \hat{\boldsymbol{r}}\hat{\boldsymbol{r}}$$

$$\xlongequal[P66(8)]{[5]} -\sqrt{3}\left[[\boldsymbol{\sigma}_l \otimes \hat{\boldsymbol{r}}]^0 \otimes \hat{\boldsymbol{r}}\right]^1 \tag{B-12}$$

$$\xlongequal[P69(1)]{[5]} -\sqrt{3}\sum_{h=0}^{2}\hat{h}\begin{Bmatrix} 1 & 1 & 0 \\ 1 & 1 & h \end{Bmatrix}\left[\boldsymbol{\sigma}_l \otimes [\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}]^h\right]^1 \tag{B-13}$$

$$\xlongequal[P299(1)]{[5]} -\sqrt{3}\sum_{h=0}^{2}\hat{h}(-)^h\frac{1}{3}\left[\boldsymbol{\sigma}_l \otimes [\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}]^h\right]^1 \tag{B-14}$$

$$= \frac{1}{\sqrt{3}}\sum_{h=0}^{2}(-)^{h+1}\hat{h}\left[\boldsymbol{\sigma}_l \otimes [\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}]^h\right]^1 \tag{B-15}$$

★ $[\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}]^h$

$$= [\hat{\boldsymbol{C}}_1 \otimes \hat{\boldsymbol{C}}_1]^h \tag{B-16}$$

$$\xlongequal[P131(7)]{[5]} \frac{4\pi}{3}[\hat{\boldsymbol{Y}}_1 \otimes \hat{\boldsymbol{Y}}_1]^h \tag{B-17}$$

$$\xlongequal[P144(14)]{[5]} \frac{4\pi}{3}\sqrt{\frac{9}{4\pi(2h+1)}}C_{1010}^{h0}\hat{\boldsymbol{Y}}_h \tag{B-18}$$

$$= \sqrt{\frac{4\pi}{2h+1}}C_{1010}^{h0}\hat{\boldsymbol{Y}}_h \tag{B-19}$$

★ $\left\langle [NL \otimes nl]^{L_{ab}} \middle\| Y_2(\sqrt{2}r)[\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}]^h \middle\| [N'L' \otimes n'l']^{L_{a'b'}} \right\rangle$

$$\xlongequal{(B-19)} \sqrt{\frac{4\pi}{2h+1}}C_{1010}^{h0}\left\langle [NL \otimes nl]^{L_{ab}} \middle\| Y_2(\sqrt{2}r)\hat{\boldsymbol{Y}}_h \middle\| [N'L' \otimes n'l']^{L_{a'b'}} \right\rangle \tag{B-20}$$

$$\xlongequal{(A-17)} \sqrt{\frac{4\pi}{2h+1}}C_{1010}^{h0}\delta_{NN'}\delta_{LL'}(-)^{h+L'+l'+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & h \\ l & l' & L' \end{Bmatrix}\left\langle nl \middle\| Y_2(\sqrt{2}r)\hat{\boldsymbol{Y}}_h \middle\| n'l' \right\rangle \tag{B-21}$$

$$= \sqrt{\frac{4\pi}{2h+1}}C_{1010}^{h0}\delta_{NN'}\delta_{LL'}(-)^{h+L'+l'+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & h \\ l & l' & L' \end{Bmatrix}\left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle\left\langle \underbrace{l}_{\hat{\boldsymbol{r}}} \middle\| \hat{\boldsymbol{Y}}_h \middle\| l' \right\rangle \tag{B-22}$$

$$\xlongequal[P496(105)]{[5]} \sqrt{\frac{4\pi}{2h+1}}C_{1010}^{h0}\delta_{NN'}\delta_{LL'}(-)^{h+L'+l'+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & h \\ l & l' & L' \end{Bmatrix}$$
$$\cdot \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle\sqrt{\frac{(2h+1)(2l'+1)}{4\pi}}C_{l'0h0}^{l0} \tag{B-23}$$

$$= \delta_{NN'}\delta_{LL'}(-)^{h+L+l'+L_{ab}}C_{1010}^{h0}C_{l'0h0}^{l0}\hat{l}'\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & h \\ l & l' & L \end{Bmatrix}\left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle \tag{B-24}$$

$$\bigstar \ \left\langle [nl\otimes NL]^{L_{ab}} \middle\| Y_2(\sqrt{2}r)[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h \middle\| [n'l'\otimes N'L']^{L_{a'b'}} \right\rangle$$

$$\overset{(B-19)}{=\!=\!=} \sqrt{\frac{4\pi}{2h+1}} C_{1010}^{h0} \left\langle [nl\otimes NL]^{L_{ab}} \middle\| Y_2(\sqrt{2}r)\hat{\mathbf{Y}}_h \middle\| [n'l'\otimes N'L']^{L_{a'b'}} \right\rangle \tag{B-25}$$

$$\overset{(A-14)}{=\!=\!=} \sqrt{\frac{4\pi}{2h+1}} C_{1010}^{h0} \delta_{NN'}\delta_{LL'}(-)^{h+l+L'+L_{a'b'}} \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix} \left\langle \underbrace{nl}_{\boldsymbol{r}} \middle\| Y_2(\sqrt{2}r)\hat{\mathbf{Y}}_h \middle\| n'l' \right\rangle \tag{B-26}$$

$$= \sqrt{\frac{4\pi}{2h+1}} C_{1010}^{h0} \delta_{NN'}\delta_{LL'}(-)^{h+l+L'+L_{a'b'}} \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle \left\langle \underbrace{l}_{\hat{\boldsymbol{r}}} \middle\| \hat{\mathbf{Y}}_h \middle\| l' \right\rangle \tag{B-27}$$

$$\overset{[5]}{\underset{P496(105)}{=\!=\!=}} \sqrt{\frac{4\pi}{2h+1}} C_{1010}^{h0} \delta_{NN'}\delta_{LL'}(-)^{h+l+L'+L_{a'b'}} \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix}$$

$$\cdot \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle \sqrt{\frac{(2h+1)(2l'+1)}{4\pi}} C_{l'0h0}^{l0} \tag{B-28}$$

$$= \delta_{NN'}\delta_{LL'}(-)^{h+l+L'+L_{a'b'}} C_{1010}^{h0} C_{l'0h0}^{l0} \hat{l}' \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} L_{ab} & L_{a'b'} & h \\ l' & l & L' \end{Bmatrix} \left\langle \underbrace{nl}_{r} \middle| Y_2(\sqrt{2}r) \middle| n'l' \right\rangle \tag{B-29}$$

$$\bigstar \ \left\langle [n_a l_a\otimes n_b l_b]^{L_{ab}} \middle\| \delta(\boldsymbol{r}_1-\boldsymbol{r}_2) \middle\| [n_{a'}l_{a'}\otimes n_{b'}l_{b'}]^{L_{a'b'}} \right\rangle$$

$$\overset{(A-19)}{=\!=\!=} \sum_{l=0}^{\infty} (-)^l \, \hat{l} \left\langle [n_a l_a\otimes n_b l_b]^{L_{ab}} \middle\| \frac{\delta(r_1-r_2)}{r_1^2} \left[\hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1)\otimes\hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2)\right]^0 \middle\| [n_{a'}l_{a'}\otimes n_{b'}l_{b'}]^{L_{a'b'}} \right\rangle \tag{B-30}$$

$$\overset{(A-10)}{=\!=\!=} \sum_{l=0}^{\infty} (-)^l \, \hat{l} \, \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} l & l & 0 \\ l_a & l_b & L_{ab} \\ l_{a'} & l_{b'} & L_{a'b'} \end{Bmatrix} \delta(r_1-r_2) \left\langle \underbrace{n_a l_a}_{\boldsymbol{r}_1} \middle\| \frac{1}{r_1^2}\hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1) \middle\| n_{a'}l_{a'} \right\rangle$$

$$\cdot \left\langle \underbrace{n_b l_b}_{\boldsymbol{r}_2} \middle\| \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2) \middle\| n_{b'}l_{b'} \right\rangle \tag{B-31}$$

$$= \sum_{l=0}^{\infty} (-)^l \, \hat{l} \, \hat{L}_{ab}\hat{L}_{a'b'} \begin{Bmatrix} l & l & 0 \\ l_a & l_b & L_{ab} \\ l_{a'} & l_{b'} & L_{a'b'} \end{Bmatrix} \delta(r_1-r_2) \left\langle \underbrace{n_a l_a}_{r_1} \middle| \frac{1}{r_1^2} \middle| n_{a'}l_{a'} \right\rangle \left\langle \underbrace{l_a}_{\hat{\boldsymbol{r}}_1} \middle\| \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1) \middle\| l_{a'} \right\rangle$$

$$\cdot \left\langle \underbrace{n_b l_b}_{r_2} \middle| 1 \middle| n_{b'}l_{b'} \right\rangle \left\langle \underbrace{l_b}_{\hat{\boldsymbol{r}}_2} \middle\| \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2) \middle\| l_{b'} \right\rangle \tag{B-32}$$

$$= \sum_{l=0}^{\infty} (-)^l \, \hat{l} \, \hat{L}_{ab} \hat{L}_{a'b'} \begin{Bmatrix} l & l & 0 \\ l_a & l_b & L_{ab} \\ l_{a'} & l_{b'} & L_{a'b'} \end{Bmatrix} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle \Big\langle \underbrace{l_a}_{\hat{\boldsymbol{r}}_1} \Big\| \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_1) \Big\| l_{a'} \Big\rangle \Big\langle \underbrace{l_b}_{\hat{\boldsymbol{r}}_2} \Big\| \hat{\mathbf{Y}}_l(\hat{\boldsymbol{r}}_2) \Big\| l_{b'} \Big\rangle \tag{B-33}$$

$$\xlongequal[P496(105)]{[5]} \sum_{l=0}^{\infty} (-)^l \, \hat{l} \, \hat{L}_{ab} \hat{L}_{a'b'} \begin{Bmatrix} l & l & 0 \\ l_a & l_b & L_{ab} \\ l_{a'} & l_{b'} & L_{a'b'} \end{Bmatrix} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle$$

$$\cdot \sqrt{\frac{(2l_{a'}+1)(2l+1)}{4\pi}} \, C^{l_a 0}_{l_{a'} 0 l 0} \sqrt{\frac{(2l_{b'}+1)(2l+1)}{4\pi}} \, C^{l_b 0}_{l_{b'} 0 l 0} \tag{B-34}$$

$$\xlongequal[P357(2)]{[5]} \sum_{l=0}^{\infty} (-)^l \, \hat{l} \, \hat{L}^2_{ab} \delta_{L_{ab} L_{a'b'}} (-)^{l+l_{a'}+l_b+L_{ab}} \frac{1}{\hat{l} \hat{L}_{ab}} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle$$

$$\cdot \sqrt{\frac{(2l_{a'}+1)(2l+1)}{4\pi}} \, C^{l_a 0}_{l_{a'} 0 l 0} \sqrt{\frac{(2l_{b'}+1)(2l+1)}{4\pi}} \, C^{l_b 0}_{l_{b'} 0 l 0} \tag{B-35}$$

$$= \sum_{l=0}^{\infty} (-)^{l_{a'}+l_b+L_{ab}} \frac{1}{4\pi} \delta_{L_{ab} L_{a'b'}} \, \hat{l}^2 \, \hat{l}_{a'} \hat{l}_{b'} \hat{L}_{ab} \, C^{l_a 0}_{l_{a'} 0 l 0} \, C^{l_b 0}_{l_{b'} 0 l 0} \begin{Bmatrix} l_{b'} & l_{a'} & L_{ab} \\ l_a & l_b & l \end{Bmatrix} \Big\langle \underbrace{n_a l_a}_{r_1} \underbrace{n_b l_b}_{r_1} \Big| n_{a'} l_{a'} n_{b'} l_{b'} \Big\rangle \tag{B-36}$$

$$\bigstar \; \big\langle \tau_a \tau_b \big| \boldsymbol{\tau}^-_{\times} \big| \tau_{a'} \tau_{b'} \big\rangle = \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^- \big| \tau_{a'} \tau_{b'} \big\rangle$$

$$\xlongequal[P5(19)]{[3]} \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^x - i (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^y \big| \tau_{a'} \tau_{b'} \big\rangle \tag{B-37}$$

$$\xlongequal[P68(30)]{[5]} \sqrt{2} \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^{1-1} \big| \tau_{a'} \tau_{b'} \big\rangle \tag{B-38}$$

$$\xlongequal[P66(5)]{[5]} -2i \big\langle \tau_a \tau_b \big| [\boldsymbol{\tau}_k \otimes \boldsymbol{\tau}_l]^{1-1} \big| \tau_{a'} \tau_{b'} \big\rangle \tag{B-39}$$

$$\xlongequal[P479(25)]{[5]} -i \sum_{\alpha\beta} C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'} 1\alpha} C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'} 1\beta} C^{1-1}_{1\alpha 1\beta} \Big\langle \frac{1}{2} \Big\| \boldsymbol{\tau}_k \Big\| \frac{1}{2} \Big\rangle \Big\langle \frac{1}{2} \Big\| \boldsymbol{\tau}_l \Big\| \frac{1}{2} \Big\rangle \tag{B-40}$$

$$= -6i \sum_{\alpha\beta} C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'} 1\alpha} C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'} 1\beta} C^{1-1}_{1\alpha 1\beta} \tag{B-41}$$

$$= -6i \Big(C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'} 1-1} C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'} 10} C^{1-1}_{1-110} + C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'} 10} C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'} 1-1} C^{1-1}_{101-1} \Big) \tag{B-42}$$

$$= 4i \Big(\delta_{\tau_a, -\frac{1}{2}} \delta_{\tau_{a'}, \frac{1}{2}} \delta_{\tau_b, \tau_{b'}} \tau_b - \delta_{\tau_b, -\frac{1}{2}} \delta_{\tau_{b'}, \frac{1}{2}} \delta_{\tau_a, \tau_{a'}} \tau_a \Big) \tag{B-43}$$

$$\bigstar \; \big\langle \tau_a \tau_b \big| \boldsymbol{\tau}^+_{\times} \big| \tau_{a'} \tau_{b'} \big\rangle = \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^+ \big| \tau_{a'} \tau_{b'} \big\rangle$$

$$\xlongequal[P5(19)]{[3]} \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^x + i (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^y \big| \tau_{a'} \tau_{b'} \big\rangle \tag{B-44}$$

$$\xlongequal[P68(30)]{[5]} -\sqrt{2} \big\langle \tau_a \tau_b \big| (\boldsymbol{\tau}_k \times \boldsymbol{\tau}_l)^{1+1} \big| \tau_{a'} \tau_{b'} \big\rangle \tag{B-45}$$

$$\xlongequal[P66(5)]{[5]} 2i\left\langle\tau_a\tau_b\middle|[\boldsymbol{\tau}_k\otimes\boldsymbol{\tau}_l]^{1+1}\middle|\tau_{a'}\tau_{b'}\right\rangle \tag{B-46}$$

$$\xlongequal[P479(25)]{[5]} i\sum_{\alpha\beta}C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'}1\alpha}C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'}1\beta}C^{11}_{1\alpha1\beta}\left\langle\frac{1}{2}\middle\|\boldsymbol{\tau}_k\middle\|\frac{1}{2}\right\rangle\left\langle\frac{1}{2}\middle\|\boldsymbol{\tau}_l\middle\|\frac{1}{2}\right\rangle \tag{B-47}$$

$$= 6i\sum_{\alpha\beta}C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'}1\alpha}C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'}1\beta}C^{11}_{1\alpha1\beta} \tag{B-48}$$

$$= 6i\Big(C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'}11}C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'}10}C^{11}_{1110}+C^{\frac{1}{2}\tau_a}_{\frac{1}{2}\tau_{a'}10}C^{\frac{1}{2}\tau_b}_{\frac{1}{2}\tau_{b'}11}C^{11}_{1011}\Big) \tag{B-49}$$

$$= 4i\Big(-\delta_{\tau_a,\frac{1}{2}}\delta_{\tau_{a'},-\frac{1}{2}}\delta_{\tau_b,\tau_{b'}}\tau_b+\delta_{\tau_b,\frac{1}{2}}\delta_{\tau_{b'},-\frac{1}{2}}\delta_{\tau_a,\tau_{a'}}\tau_a\Big) \tag{B-50}$$

★ $$\left\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\middle\|\boldsymbol{\sigma}_\times\middle\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle=\left\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\middle\|\boldsymbol{\sigma}_k\times\boldsymbol{\sigma}_l\middle\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle$$

$$\xlongequal[P66(4)]{[5]} -\sqrt{2}i\left\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\middle\|[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^1\middle\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle \tag{B-51}$$

$$\xlongequal{(A-10)} -\sqrt{2}i\sqrt{3}\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix}1&1&1\\ \frac{1}{2}&\frac{1}{2}&S_{ab}\\ \frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix}\left\langle\frac{1}{2}\middle\|\boldsymbol{\sigma}_k\middle\|\frac{1}{2}\right\rangle\left\langle\frac{1}{2}\middle\|\boldsymbol{\sigma}_l\middle\|\frac{1}{2}\right\rangle \tag{B-52}$$

$$\xlongequal{(A-3)} -6\sqrt{6}i\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix}1&1&1\\ \frac{1}{2}&\frac{1}{2}&S_{ab}\\ \frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix} \tag{B-53}$$

★ $$\boldsymbol{\sigma}_k\times\hat{\boldsymbol{r}}\,\boldsymbol{\sigma}_l\cdot\hat{\boldsymbol{r}} \tag{B-54}$$

$$\xlongequal[P66(13)]{[5]} \sqrt{6}i\big[[\boldsymbol{\sigma}_k\otimes\hat{\boldsymbol{r}}]^1\otimes[\boldsymbol{\sigma}_l\otimes\hat{\boldsymbol{r}}]^0\big]^1 \tag{B-55}$$

$$\xlongequal[P70(11)]{[5]} \sqrt{6}i\sum_{g,h=0}^{2}\sqrt{3}\hat{g}\hat{h}\begin{Bmatrix}1&1&1\\1&1&0\\g&h&1\end{Bmatrix}\big[[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^g\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\big]^1 \tag{B-56}$$

$$\xlongequal[P357(2)]{[5]} \sqrt{6}i\sum_{g,h=0}^{2}\sqrt{3}\,\hat{g}\,\hat{h}\,(-)^{h+1}\frac{1}{3}\begin{Bmatrix}1&1&1\\g&h&1\end{Bmatrix}\big[[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^g\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\big]^1 \tag{B-57}$$

$$= \sqrt{2}i\sum_{g,h=0}^{2}\hat{g}\,\hat{h}\,(-)^{h+1}\begin{Bmatrix}1&1&1\\g&h&1\end{Bmatrix}\big[[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^g\otimes[\hat{\boldsymbol{r}}\otimes\hat{\boldsymbol{r}}]^h\big]^1 \tag{B-58}$$

★ $$\left\langle[\frac{1}{2}\otimes\frac{1}{2}]^{S_{ab}}\middle\|[\boldsymbol{\sigma}_k\otimes\boldsymbol{\sigma}_l]^g\middle\|[\frac{1}{2}\otimes\frac{1}{2}]^{S_{a'b'}}\right\rangle \tag{B-59}$$

$$\xlongequal{A-10} \hat{g}\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix}1&1&g\\ \frac{1}{2}&\frac{1}{2}&S_{ab}\\ \frac{1}{2}&\frac{1}{2}&S_{a'b'}\end{Bmatrix}\left\langle\frac{1}{2}\middle\|\boldsymbol{\sigma}_k\middle\|\frac{1}{2}\right\rangle\left\langle\frac{1}{2}\middle\|\boldsymbol{\sigma}_l\middle\|\frac{1}{2}\right\rangle \tag{B-60}$$

$$\overset{A-3}{=\!=} 6\hat{g}\hat{S}_{ab}\hat{S}_{a'b'}\begin{Bmatrix} 1 & 1 & g \\ \frac{1}{2} & \frac{1}{2} & S_{ab} \\ \frac{1}{2} & \frac{1}{2} & S_{a'b'} \end{Bmatrix} \tag{B-61}$$

★ $\left\langle [nl \otimes NL]^{L_{ab}} \middle\| r\hat{\boldsymbol{r}} \middle\| [n'l' \otimes N'L']^{L_{a'b'}} \right\rangle$

$$= \delta_{NN'}\delta_{LL'}(-)^{1+l+L'+L_{a'b'}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{ab} & L_{a'b'} & 1 \\ l' & l & L' \end{Bmatrix}\left\langle \underbrace{nl}_{r} \middle| r \middle| n'l' \right\rangle \left\langle \underbrace{l}_{\hat{\boldsymbol{r}}} \middle\| \hat{\boldsymbol{r}} \middle\| l' \right\rangle \tag{B-62}$$

$$= \delta_{NN'}\delta_{LL'}(-)^{1+l+L'+L_{a'b'}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{ab} & L_{a'b'} & 1 \\ l' & l & L' \end{Bmatrix}\left\langle \underbrace{nl}_{r} \middle| r \middle| n'l' \right\rangle \hat{l}' C^{l0}_{l'010} \tag{B-63}$$

★ $\left\langle [n_a l_a \otimes n_b l_b]^{L_{ab}} \middle\| r_k\hat{\boldsymbol{r}}_k \middle\| [n_{a'} l_{a'} \otimes n_{b'} l_{b'}]^{L_{a'b'}} \right\rangle$

$$= \delta_{n_a n_{a'}}\delta_{l_a l_{a'}}(-)^{1+l_{a'}+l_{b'}+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & 1 \\ l_b & l_{b'} & l_{a'} \end{Bmatrix}\left\langle n_b l_b \middle\| r_k\hat{\boldsymbol{r}}_k \middle\| n_{b'} l_{b'} \right\rangle \tag{B-64}$$

$$= \delta_{n_a n_{a'}}\delta_{l_a l_{a'}}(-)^{1+l_{a'}+l_{b'}+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & 1 \\ l_b & l_{b'} & l_{a'} \end{Bmatrix}\left\langle n_b l_b \middle| r_k \middle| n_{b'} l_{b'} \right\rangle \left\langle l_b \middle\| \hat{\boldsymbol{r}}_k \middle\| l_{b'} \right\rangle \tag{B-65}$$

$$= \delta_{n_a n_{a'}}\delta_{l_a l_{a'}}(-)^{1+l_{a'}+l_{b'}+L_{ab}}\hat{L}_{ab}\hat{L}_{a'b'}\begin{Bmatrix} L_{a'b'} & L_{ab} & 1 \\ l_b & l_{b'} & l_{a'} \end{Bmatrix}\left\langle n_b l_b \middle| r_k \middle| n_{b'} l_{b'} \right\rangle \hat{l_{b'}} C^{l_b 0}_{l_{b'}010} \tag{B-66}$$

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